

TOPOLOGICAL CATEGORICITY OF EFFECTIVE POLISH SPACES

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ABSTRACT. In this paper we introduce the notion of topological computable categoricity for a Polish space. We show that, in contrast to computable structures, homogeneous Polish spaces tend not to be computably categorical. In particular, the infinite countable discrete space, the Baire space, and the Hilbert cube are all not topologically computably categorical. We also show that spaces which are computably compact or have a presentation with a computable base of open sets similarly fail to be topologically computably categorical. Finally, we investigate the degrees of categoricity for the Baire space and the closed unit interval (the arc).

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1. INTRODUCTION

In this paper, we study the algorithmic aspects of topological spaces. This contributes to a general research paradigm in computability theory that investigates the relationship between algorithmic aspects and structural properties of a given structure. One line of investigation within this paradigm

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that has captured the interest of many researchers is the study of *computable structures*, which are structures with domain ω and whose operations and relations are uniformly computable [Rab60, Mal61]. For a fixed (countable) structure $\mathcal{A}$, we refer to a computable structure $C \cong \mathcal{A}$ as a *computable presentation* of $\mathcal{A}$.

One natural approach to studying the algorithmic aspects of countable structures is to investigate which structures have computable presentations, i.e., are *computable*. Clearly, there are at most countably many computable presentations, so not every countable group or linear order will have a computable presentation. On the other hand, most (if not all) countable algebraic structures that occur in everyday mathematical practice are computably presented. In fact, it takes some effort to construct explicit examples of structures that are not computably presented (see [Fei70, Khi98]). In most such early diagonalisation arguments, the resulting structure is not too far from being computably presented, in the sense that the halting problem $0'$ can usually compute a presentation of such a structure. But of course, this approach can be iterated further to show that there are structures computable in $0''$ – the second iteration of the halting problem – but with no $0'$ -computable isomorphic copies, and so on. Expanding upon this approach, one may attempt to measure how non-computable it is to produce a presentation of $\mathcal{A}$ (see, for instance, [Ric81, Kni86, AJK90, JS91, DK92, JS94, Mel13b]).

Of course, it is perhaps most natural to view computable presentations up to computable isomorphism – an idea that can be traced back to [Rab60, Mal61]. For example, it is easy to see that, for any two given computable presentations of the countable dense linear order $(\mathbb{Q}, <)$, the standard back-and-forth argument produces a computable isomorphism between them. In fact, any suitably homogeneous computable structure will also have this property. We call such structures – i.e., those with exactly one computable presentation up to computable isomorphism – *computably categorical*. In contrast with the situation for computable presentability, it seems that many structures commonly encountered in mathematical practice are *not* computably categorical, apart from the finitely generated and homogeneous ones. In a similar vein as before, for structures that are not computably categorical, one can then investigate the algorithmic content encoded by isomorphisms between presentations of the given structure. To this end, the complexity of isomorphisms between (computable) presentations of a structure has been studied extensively (see, for instance, [McC02, McC03, Mil09, MS13, CHR14, FZ25]). The ‘reverse’ question, which, given a Turing degree $\mathbf{d}$, asks whether there are structures that are ‘strictly $\mathbf{d}$ -categorical’ (to be made precise later), has also garnered considerable attention [FKM10, AC12, CFS13, FS14, CDHTM20, CR24].

However, as mentioned at the beginning of the paper, the primary objects of interest in the present article are topological spaces, which are generally not countable. Nonetheless, one can similarly define what it means for an uncountable space to be computably presented. In this article, we shall be primarily interested in studying *computable Polish spaces*; these are Polish spaces $\mathcal{X} \cong (S, d)$, where $S = \{x_0, x_1, \dots\}$ is a countable set and $d(x_i, x_j)$ is a computable real uniformly in i, j [Lac57, Mos64]. Despite the obvious analogy with computable structure theory, most of the basic notions of effective presentability for Polish spaces have only recently been studied up to homeomorphism [GKP17, HTMN20, BHTM23, LMN23, KMN25]. Just as for computable structures, for a Polish space that is not computable, researchers have also begun studying its *degree spectrum*, the collection of Turing degrees that produce presentations of the given Polish space [Sel20, Mel21, HKS23].

In this article, rather than studying the algorithmic content encoded in the isomorphism type of Polish spaces, we instead focus on the algorithmic content of homeomorphisms between computable

presentations. We adopt a Type 2 approach (see [PER89, Wei00]) when considering maps between computable Polish spaces.

Definition 1.1. Let $(\{x_i\}_{i \in \mathbb{N}}, d_0), (\{y_i\}_{i \in \mathbb{N}}, d_1)$ be computable presentations of Polish spaces $\mathcal{X}, \mathcal{Y}$ respectively. $f : \mathcal{X} \rightarrow \mathcal{Y}$ is *sequentially computable* if there is a Turing functional Φ such that for each fast Cauchy sequence $\{x_{i_j}\}_{j \in \mathbb{N}}$ (i.e., $d_0(x_{i_j}, x_{i_{j+1}}) < 2^{-j}$ for each $j \in \mathbb{N}$), $\Phi(\{x_{i_j}\}_{j \in \mathbb{N}})$ produces a fast Cauchy sequence $\{y_{i_j}\}_{j \in \mathbb{N}}$ converging to $f(\lim_j x_{i_j})$.

It is easy to see that for computable Polish spaces $\mathcal{X}, \mathcal{Y}$, a function $f : \mathcal{X} \rightarrow \mathcal{Y}$ is sequentially computable iff it is effectively continuous [Cei59] (also see [DM23, Lemma 2.13]):

Definition 1.2. In the notation of the previous definition, $f : \mathcal{X} \rightarrow \mathcal{Y}$ is *effectively continuous* if there is a c.e. set W so that for any $i \in \mathbb{N}$ and $r \in \mathbb{Q}^+$,

$$f^{-1}(B^{\mathcal{Y}}(y_i, r)) = \bigcup_{\langle i, r, j, q \rangle \in W} B^{\mathcal{X}}(x_j, q),$$

where $B^{\mathcal{M}}(x, q)$ stands for the basic open ball in space $\mathcal{M}$ centred in $x \in \mathcal{M}$ and of radius $q \in \mathbb{Q}$.

In view of the equivalence of these two definitions in the context of computable Polish spaces, we shall refer to sequentially computable functions as simply *computable*. It is also not too hard to see that the inverse of a bijective computable function need not be sequentially computable, even if the function is indeed (classically) a homomorphism. Intuitively, this is because a sequentially computable map may send ‘complicated’ fast Cauchy sequences to ‘simple’ ones, thereby causing its inverse to fail to be sequentially computable. This differs from the Type 1 setting that occurs in countable structures, where the inverse of a bijective computable function is necessarily computable. Similarly, the inverse of a surjective isometry is also easily seen to be computable. This gives rise to an interesting approach in which “being homeomorphic” can be treated as a reduction rather than an equivalence [GMST, KKM⁺a, KKM⁺b].

In this article we shall restrict ourselves to homeomorphisms f such that both f and f^{-1} are effectively continuous, and study computable Polish spaces up to such maps, which we will call *bi-computable homeomorphisms*. Following the analogy with computable structures, we arrive at the following:

Definition 1.3. A computable Polish space is *topologically computably categorical* if it has a unique computable Polish presentation, up to bi-computable homeomorphism.

We will drop “topologically” throughout the paper and refer to these simply as computably categorical spaces, since there is no danger of confusion with, say, isometric computable categoricity (which is also called computably categorical in the literature [Mel13a, Ilj10]). We also note that a different approach was taken in [DM26, BHTM23], where the notion was restricted to computably compact presentations (Def. 2.3). Such (compact) spaces were called effectively categorical [BHTM23] or computably categorical [DM26]. Our goal is to have a definition that works for spaces that are not even compact, let alone computably compact. However, the reader should keep in mind that there is a potential conflict of terminology with [DM26]. It also follows from the results of the paper that 2^ω is not topologically computably categorical, but it is effectively categorical [BHTM23]. Thus the notions are not equivalent even for the standard computably compact spaces. We could of course say “topologically computably categorical with respect to computable Polish presentations” to distinguish our approach from that taken in [DM26] and [Mel13a, Ilj10], but it would make the term way too long. We will also drop “bi-” in “bi-computable homeomorphism” and refer to them simply as “computable homeomorphisms”, but we shall always keep in mind that

they must be computable both ways. The reason is that we will later consider $\mathbf{d}$ -bi-computable isomorphisms, which would be a suboptimal choice of terminology. We hope that there is no danger of confusion.

A question that immediately comes to mind is as follows:

Question 1.4. *Is there an infinite computable Polish space which is computably categorical?*

Applying the intuition from computable structure theory, a natural conjecture would be that some suitably homogeneous Polish space—say, for instance, the countable discrete Polish space, the Cantor space, the Baire space, or the Hilbert cube—should be computably categorical. However, we show the following.

Theorem 1.5. *The following Polish spaces are not (topologically) computably categorical:*

- (1) *countable Polish spaces;*
- (2) *the Baire space;*
- (3) *computably compact spaces, which include the Hilbert cube and 2^ω ;*
- (4) *spaces that admit a computable Polish presentation where every basic open ball is connected.*

Different items of the theorem will appear as separate theorems throughout the paper: (1) is Theorem 3.4, (2) is implied by the stronger Theorem 4.8, (3) follows from Theorem 5.1 and a folklore result (Lemma 2.5), and (4) reappears as Theorem 6.1. The reader should note that the last two items of the theorem cover very broad classes of spaces. For instance, it takes quite a bit of effort to come up with an example of a (compact) computable Polish space that is not homeomorphic to any computably compact space; while such examples do exist ([HTMN20, HKS23, DM23, BBG⁺25, KMN25]), all known proofs require some relatively sophisticated machinery and result in rather pathological spaces. Thus, all common compact Polish spaces in the literature will be *not* computably categorical, according to (3) of the theorem. On the other hand, (4) will cover various non-compact locally connected spaces, such as separable Hilbert spaces, which are not even necessarily locally compact.

Different items of the theorem will require substantially different techniques, which may indicate that Question 1.4 is much more subtle than it may appear. It is worth noting that, in computable structure theory, computable categoricity is not exclusively witnessed by homogeneous or finitely generated structures, and indeed computably categorical structures can be extremely intricate; see, e.g., [MS13]. Thus, it is conceivable that analogous techniques or ideas could be applied to construct an infinite computably categorical space. Thus, we suspect that the answer to Question 1.4 may be positive despite the partial evidence provided by Theorem 1.5.

Since (quite frustratingly) the only examples of computably categorical spaces that we know of are finite, and since all common Polish spaces appear not to be computably categorical, we shall look beyond computable categoricity. For computable Polish spaces $\mathcal{X}$ that are not computably categorical, one can then ask whether there is some minimal amount of noncomputable information required to produce homeomorphisms between any two given computable presentations of $\mathcal{X}$. We make this more precise in the following definitions.

Definition 1.6. A computable Polish space $\mathcal{X}$ is $\mathbf{d}$ -categorical if for any two computable presentations of $\mathcal{X}$, there is a (bi-) $\mathbf{d}$ -computable homeomorphism f between them¹.

Following the analogy with computable structures [FKM10], we arrive at

¹This means that both f and f^{-1} are $\mathbf{d}$ -effectively continuous.

Definition 1.7. A computable Polish space $\mathcal{X}$ has *degree of (topological) categoricity* $\mathbf{d}$ if $\mathbf{d}$ is the least Turing degree such that $\mathcal{X}$ is (topologically) $\mathbf{d}$ -categorical (if it exists).

In this paper, we also study the degrees of categoricity for various common Polish spaces, including countable spaces, the Baire space, and the arc (the unit interval). Some of these spaces turn out to be much harder to handle than we expected, and especially the Baire space. Indeed, these spaces may be well understood topologically, but when it comes to dynamic approximations of spaces and maps between them, these classical results are not particularly helpful.

We subdivide the paper into several sections, each devoted to a space or class of spaces. Our investigation of each individual space or class of spaces usually goes beyond the techniques and results necessary to establish the corresponding item in Theorem 1.5.

After a short preliminaries Section 2, in Section 3 we prove several results about countable Polish spaces, including Theorem 3.4, which is also item (1) of Theorem 1.5. In Section 4 we prove the already mentioned Theorem 4.8, which states that the Baire space is not $\mathbf{0}'$ -categorical. Section 4 contains further results about ω^ω , specifically those that give a more satisfactory structural description of categoricity when restricted to presentations of the Baire space with a natural additional property; we postpone their statement until Section 4.

Sections 5 and 6 contain detailed proofs of items (3) and (4) of Theorem 1.5, respectively. Of course, the most basic example of a space satisfying the assumptions of (3) and (4) is the arc, or the unit closed interval $[0, 1]$. Section 7 concludes the paper with a positive result, Theorem 7.1, which establishes that the degree of categoricity of the arc is $\mathbf{0}''$.

It appears that in all examples known to us so far, if a degree of topological categoricity exists, it is $\geq \mathbf{0}'$. Thus, we also leave open the following question:

Question 1.8. Which Turing degrees apart from $\mathbf{0}^{(\alpha)}$ for a computable ordinal α are degrees of (topological) categoricity for a computable Polish space? In fact, is there an infinite computable Polish space that is (topologically) $\mathbf{d}$ -categorical for some $\mathbf{d} \not\geq \mathbf{0}'$?

2. PRELIMINARIES

Recall that a real number x is *computable* [Tur37] if there is a Turing machine such that given $n \in \omega$ outputs a rational number r such that $|x - r| < 2^{-n}$.

Definition 2.1. Given a Polish space $\mathcal{M} \cong (\overline{\{\alpha_i\}_{i \in \omega}}, d)$, the standard topological basis is given by the collection of open balls $\{B(\alpha_i, r)\}_{i \in \omega, r \in \mathbb{Q}}$, where $B(\alpha_i, r) = \{x \in \mathcal{M} \mid d(x, \alpha_i) < r\}$. We call this a *basic open ball*, or simply an open ball. The closed ball $\overline{B}(\alpha_i, r)$ is $\{x \in \mathcal{M} \mid d(x, \alpha_i) \leq r\}$. Whenever convenient, we will abbreviate these as B_r^i and $\overline{B}_r^i$ respectively.

Given any $x \in \mathcal{M}$, a *name* for x is any $p \in \omega^\omega$ such that the sequence $\{\alpha_{p(i)}\}_{i \in \omega}$ is fast converging (i.e. $d(\alpha_i, \alpha_j) < 2^{-\max\{i, j\}}$ for every i, j) and converges to x . An *effective open set* $\mathcal{O} \subseteq \mathcal{M}$ is one where $\mathcal{O} = \bigcup_{(i, r) \in W} B_r^i$ for some c.e. set W . A subset of $\mathcal{M}$ is *effectively closed* if its complement is

effectively open. We denote $\bigcup_{(i, r) \in W} B_r^i$ by $\mathcal{O}(W)$.

We remark here that the collection of basic open balls depend on the computable Polish presentation, rather than the isomorphism type of the Polish space.

Definition 2.2 (Formal predicates). Fix a computable Polish presentation $(\{\alpha_n\}_{n \in \omega}, d)$. Given two open balls B_r^i and B_q^j , we say that B_r^i is *formally contained* in B_q^j , written as $B_r^i \subseteq^f B_q^j$,

if $d(\alpha_i, \alpha_j) + r < q$. The formal containment relation is c.e. (given the indices of open balls) while containment of open balls ($B_r^i \subseteq B_q^j$) is generally Π_1^1 . In many cases, working with formal containment is enough.

Given an open (or closed) ball B_p^i , the radius of the ball is denoted as $r(B_p^i) = p$. Given an open set $\bigcup_n B_{p_n}^{i_n}$, the formal diameter of $\bigcup_n B_{p_n}^{i_n}$, $\text{diam}(\bigcup_n B_{p_n}^{i_n}) = \sup_{k,m} \{d(\alpha_{i_k}, \alpha_{i_m}) + p_k + p_m\}$. The term formal obviously comes from the fact that the radius of each ball in the open set is considered and therefore depends on the given name of the open set. Obviously if U is effectively open then $\text{diam}(U)$ is left-c.e..

Definition 2.3. [Computable compactness] Given a Polish structure $\mathcal{M}$, a *finite open cover* of $\overline{\mathcal{M}}$ is a finite collection of open balls $B_0, B_1, \dots, B_n$ such that $\bigcup_{i \leq n} B_i = \overline{\mathcal{M}}$. A finite open cover can obviously be encoded by an integer, which we also denote as $\langle B_0, B_1, \dots, B_n \rangle$. An ε -*net* of $\overline{\mathcal{M}}$ is a finite open cover of $\overline{\mathcal{M}}$ where each open ball in the cover has radius less than ε .

A computable Polish structure $\mathcal{M}$ is said to be *computably compact* if one of the following equivalent statements hold:

- (i) The set of (codes for) finite open covers of $\overline{\mathcal{M}}$ is c.e.
- (ii) There is a computable function f with the property that it maps each natural number n to some 2^{-n} -net of $\overline{\mathcal{M}}$, and that if $n < m$ then every open ball in $f(m)$ is formally included in some ball of $f(n)$.

A compact Polish space is said to be *computably compact* if it has a computable Polish presentation that is computably compact. There is no reason why every computable Polish presentation of an computably compact Polish space has to be computably compact. In fact, we will later show that this is not the case (Theorem 5.1). The definitions above relativize to any oracle D . By identifying compact spaces with closed subsets of the Hilbert cube, some authors have used the notation $\mathcal{V}([0, 1]^\omega)$, the hyperspace of compact subsets of the Hilbert cube endowed with the Vietoris topology. computably compact Polish spaces are just the computable elements of $\mathcal{V}([0, 1]^\omega)$.

The following is folklore:

Lemma 2.4. *Let $\mathcal{M}$ be a compact computable Polish structure. Then $\mathcal{M}$ is $\emptyset'$ -computably compact.*

Proof. Given $n \in \omega$ we search (recursively in $\emptyset'$) for some finite collection of open balls $B_{q_0}^{i_0}, B_{q_1}^{i_1}, \dots, B_{q_N}^{i_N}$ such that the radius of each ball $q_j < 2^{-n-1}$ and every special point is in $\bigcup_{j \leq N} \overline{B_{q_j}^{i_j}}$. Then $\langle B_{2q_0}^{i_0}, B_{2q_1}^{i_1}, \dots, B_{2q_N}^{i_N} \rangle$ is a 2^{-n} -net of $\overline{\mathcal{M}}$. $\square$

The following shall also be useful in our analysis:

Lemma 2.5. *Suppose that $\mathcal{M}$ and $\mathcal{N}$ are computable Polish structures that are computably homeomorphic, and $\mathcal{M}$ is computably compact. Then $\mathcal{N}$ is computably compact as well.*

Proof. Fix a computable homeomorphism $f : \mathcal{M} \rightarrow \mathcal{N}$. For each n we search for some finite sequence of open balls $\langle B_0, \dots, B_N \rangle$ such that $r(B_i) < 2^{-n}$ for all $i \leq N$, and $\bigcup_{i \leq N} f^{-1}(B_i) = \mathcal{M}$. Then $\langle B_0, \dots, B_N \rangle$ is a 2^{-n} -net of $\mathcal{N}$ and can be found computably. $\square$

3. COUNTABLE SPACES AND CANTOR-BENDIXON RANK

We start our investigation with an easy example, the infinite countable discrete space $\mathcal{X} = (\{\alpha_i\}_{i \in \omega}, d)$, where $d(\alpha_i, \alpha_j) = 1$ for every $i \neq j$. One might be tempted to guess that the discrete space can provide an easy answer to Question 1.4. However, the following proposition, albeit being little more than just an observation, suggests the potential depth of the investigations in this paper:

Proposition 3.1. *The infinite countable discrete space has degree of categoricity $\mathbf{0}'$.*

Proof. Given any two computable Polish presentations $\mathcal{M} = (\{\alpha_i\}_{i \in \omega}, d)$ and $\mathcal{N} = (\{\alpha'_i\}_{i \in \omega}, d')$ of the infinite countable discrete space, we enumerate the ball (B_r^i, B_q^i) into W iff for every $j \neq i$, $\alpha_j \notin B_r^i$ and $\alpha'_j \notin B_q^i$. Then it is easy to see that W and W^{-1} are $\emptyset'$ -computable names for the homeomorphism $f(\alpha_i) = \alpha'_i$ and its inverse respectively.

Now we wish to encode $\emptyset'$ into a pair of Polish presentations $\mathcal{M} = (\{\alpha_i\}_{i \in \omega}, d)$ and $\mathcal{N} = (\{\alpha'_i\}_{i \in \omega}, d')$ of the infinite countable discrete space. Recall that this makes $\mathbf{0}'$ a *strong degree of categoricity* for the space. We take $\mathcal{N}$ to be the “standard” copy of $\mathcal{X}$, where $d'(\alpha_i, \alpha_j) = 1$ iff $i \neq j$. For each n and s , if n enters $\emptyset'$ at stage s we declare $d(\alpha_{2i+1}, \alpha_{2n}) = 2^{-s}$, where i is the least such that α_{2i+1} has not been used in this way for coding. We also set $d(\alpha_u, \alpha_v) = 1$ for all other pairs $u \neq v$. Clearly d is a computable metric and each special point of $\mathcal{M}$ is isolated, so $\mathcal{M}$ is a computable presentation of $\mathcal{X}$. Now fix a homeomorphism $f : \mathcal{M} \rightarrow \mathcal{N}$ and suppose we are given an oracle D where $f^{-1}(B)$ is uniformly D -c.e. Now given n search for some j such that $\alpha_{2n} \in f^{-1}(B_1^j)$; this search is obviously D -computable. Take s to be any number large enough so that $B_{2^{-s}}^{2n} \subseteq f^{-1}(B_1^j)$. Then $n \in \emptyset'$ if and only if $n \in \emptyset'[s]$. Thus $D \geq_T \emptyset'$. $\square$

Even though the infinite countable discrete space is not computably categorical, it still has a relatively low complexity according to its degree of categoricity; $\mathbf{0}'$ is exactly the level required to determine topological relationships between computable Polish presentations of the space. Since all countable Polish spaces of Cantor-Bendixson rank 1 are either finite or homeomorphic to the discrete space, all countable Polish spaces of rank 1 will have degree of categoricity either $\mathbf{0}$ or $\mathbf{0}'$. Naturally, we expect a relationship between the rank of a Polish space and a bound on categoricity for that space. An uncountable space might be complicated because of its perfect kernel, but we should still expect some kind of relationship between rank and categoricity for countable spaces.

Consider a countable computable Polish space of rank 2. Since verifying if a given special point is isolated is Σ_2^0 relative to the presentation, we might expect that this space is $\emptyset'''$ -categorical. An argument might be along the lines of the following: Fix a pair of computable Polish presentations $\mathcal{M}$ and $\mathcal{N}$ of a countable Polish space of rank 2. Given a ball B in either structure, it is Π_2^0 to tell if the ball contains infinitely many points and therefore contains a limit point. Hence it would be Π_3^0 to tell if a given ball has exactly one limit point. Since $(\mathcal{M})'$ is either finite or discrete, we can use the above to construct a homeomorphism between $\overline{\mathcal{M}}$ and $\overline{\mathcal{N}}$.

The argument above will work if the given space is *compact*. However, an open ball in a non-compact space can contain infinitely many points and still contain no limit point; this latter property is Π_1^1 . We will show that there is no arithmetical relationship between the rank of a countable space and an upperbound on the categoricity of the space:

Proposition 3.2. *For any computable ordinal α , there is a countable computable Polish space $\mathcal{X}$ of Cantor-Bendixson rank 2 such that if $\mathcal{X}$ is D -categorical, then $D \geq_T \emptyset^{(\alpha)}$ for any oracle D .*

Proof. Given a tree $T \subseteq \omega^{<\omega}$ we define the Polish structure $\mathcal{M}(T)$ by taking $\{a_\sigma \mid \sigma \in T\}$ as the set of special points. We set $d(a_\sigma, a_\tau) = 2^{-\min\{|\sigma|, |\tau|\}}$ if $\sigma \neq \tau$ are comparable, and $d(a_\sigma, a_\tau) = 2^{-i}$ if σ and τ are incomparable and i is the least such that $\sigma(i) \neq \tau(i)$. Then d is a metric on the set $\{a_\sigma \mid \sigma \in T\}$, and $\mathcal{M}(T)$ is computable if T is computable. Furthermore, it is easy to see that $x \in \overline{\mathcal{M}(T)}$ is a limit point if and only if $x = \lim_{j \rightarrow \infty} a_{z \upharpoonright j}$ for some infinite path $z \in [T]$.

Note that $\overline{\mathcal{M}(T)}$ is not necessarily homeomorphic to $[T]$ (as a subspace of the Baire space); every special point of $\overline{\mathcal{M}(T)}$ is isolated even when T is perfect. However, we do have $(\overline{\mathcal{M}(T)})' \cong [T]$.

We shall require the following folklore (see e.g. [HTMN20, Lemma 3.7]):

Fact 3.3. *Let R be a hyperarithmetical relation. Then there is a computable sequence of trees $T_n \subseteq \omega^{<\omega}$ such that for every n , if $n \in R$ then T_n has a single infinite path, and if $n \notin R$ then T_n contains no infinite path.*

We fix a hyperarithmetical non-computable set R and a computable sequence of trees T_n with the properties above. Let $\mathcal{N}$ be the space defined as follows. Let T be the tree defined by $T = \{n * \sigma \mid n \in \omega \text{ and } \sigma \in T_n\}$, and S be the tree $S = \{2n * 0^k * j \mid n, k, j \in \omega\} \cup \{(2n+1) * j \mid n, j \in \omega\}$. Then $\mathcal{M}(T)$ and $\mathcal{M}(S)$ are both computable Polish structures.

We claim that $\overline{\mathcal{M}(T)} \cong \overline{\mathcal{M}(S)}$. $\overline{\mathcal{M}(T)}$ is the discrete union of the subspaces $\overline{\mathcal{M}_n(T)}$ where $\mathcal{M}_n(T) = \{a_{n*\sigma} \mid \sigma \in T_n\}$. Similarly, $\overline{\mathcal{M}(S)}$ is the discrete union of the subspaces $\overline{\mathcal{M}_n(S)}$ where $\mathcal{M}_n(S) = \{a_{n*\sigma} \mid n * \sigma \in S\}$. Hence, it is enough to show that there is a bijection $h : \omega \rightarrow \omega$ such that $\overline{\mathcal{M}_n(T)} \cong \overline{\mathcal{M}_{h(n)}(S)}$ for all n . Define h recursively by the following: If T_n is well-founded, let $h(n) = 2k+1$ for the least k such that $2k+1 \notin \{h(0), \dots, h(n-1)\}$. Otherwise if T_n contains an infinite path then let $h(n) = 2k$ for the least k such that $2k \notin \{h(0), \dots, h(n-1)\}$. h is obviously bijective as R is infinite and co-infinite.

Now fix an n and we wish to argue that $\overline{\mathcal{M}_n(T)} \cong \overline{\mathcal{M}_{h(n)}(S)}$. If T_n is well-founded then $\overline{\mathcal{M}_n(T)}$ and $\overline{\mathcal{M}_{h(n)}(S)}$ do not have limit points and are therefore both homeomorphic to the infinite countable discrete space. So suppose that T_n contains a single infinite path z . Hence $\overline{\mathcal{M}_n(T)} = \mathcal{M}_n(T) \cup \{x\}$ for some limit point x . Since each special point of $\mathcal{M}_n(T)$ is isolated, $x \notin \mathcal{M}_n(T)$. Similarly, $\overline{\mathcal{M}_{h(n)}(S)} = \mathcal{M}_{h(n)}(S) \cup \{x'\}$ for some $x' \notin \mathcal{M}_{h(n)}(S)$. Define $f_n : \overline{\mathcal{M}_n(T)} \rightarrow \overline{\mathcal{M}_{h(n)}(S)}$ by the following. Set $f_n(x) = x'$ and $f_n(a_{n*(z \upharpoonright j)}) = a_{h(n)*0^j}$ for all j . Finally, for each k we fix some enumeration of the nodes in $P_{n,k}$, where $P_{n,k}$ is the set of all $\sigma \in T_n$ where k is the least such that $z(k) \neq \sigma(k)$. (WLOG we may assume that for each k , $P_{n,k}$ is infinite as this does not change the infinite paths of T_n). For each k we define $f_n(\sigma) = a_{h(n)*0^k*(j+1)}$ where σ is the j -th element in $P_{n,k}$.

Obviously f_n is bijective. We now write down a name for f_n and a name for f_n^{-1} . Let W consist of the following pairs of open balls. For each j and each $k \geq j$ we put $(B(a_{n*(z \upharpoonright j)}, 2^{-k}), B(a_{h(n)*0^j}, 2^{-k}))$ into W . Also for each k, j and $l > \max\{|z|, k\} + 1$ where σ is the j -th element of $P_{n,k}$, we put $(B(a_{n*\sigma}, 2^{-l}), B(a_{h(n)*0^k*(j+1)}, 2^{-l}))$ into W . It is straightforward to verify that W and W^{-1} are a name for f and f^{-1} respectively.

Note that we do not require the homeomorphism $\overline{\mathcal{M}(T)} \cong \overline{\mathcal{M}(S)}$ to be witnessed by a computable homeomorphism; in fact, we will argue that every homeomorphism between these two structures must enumerate R . This will be enough to prove the proposition. Let $F : \overline{\mathcal{M}(T)} \rightarrow \overline{\mathcal{M}(S)}$ be a homeomorphism which is D -computable for some oracle D . We enumerate n into V^D if we find $B(a_{2m*0^k}, 2^{-k}) \subseteq^F F(B(a_n, 1))$ for some m, k . Then $n \in R$ iff T_n has an infinite path iff $B(a_n, 1)$ contains a limit point iff $\lim_k a_{2m*0^k} \in F(B(a_n, 1))$ for some m . Thus, $V^D = R$ and R is c.e. relative to D . $\square$

We conclude this section by answering Question 1.4 in the negative for countable Polish spaces.

Theorem 3.4. *If $\mathcal{X}$ is a countable computable Polish space, then there exists $\mathcal{Y} \cong \mathcal{X}$, a computable Polish space, such that $\mathcal{Y}$ is not computably homeomorphic to $\mathcal{X}$.*

This proof has two parts to it. First we are given a computable space $\mathcal{X}$, and construct a second space $\mathcal{Y}$ by adding points to $\mathcal{X}$ in such a way that there is no computable continuous bijection from

$\mathcal{Y}$ to $\mathcal{X}$. Second, we will give a proof that these two spaces are homeomorphic, by defining a map on the isolated points that extends to a homeomorphism.

The construction of $\mathcal{Y}$ is a finite injury construction. The construction of the homeomorphism will require us to invoke choice several times, and works more generally by showing that if we have any countable Polish space (not necessarily computable) it is possible to ‘double’ all the points in a way that results in the same space as before.

3.1. Construction of $\mathcal{Y}$. Given a space $\mathcal{X} = (\{\alpha_n\}_{n \in \omega}, d_{\mathcal{X}})$. We will construct $\mathcal{Y} = (\{\gamma_n\} = \{\alpha_n\}_{n \in \omega} \sqcup \{\beta_n\}_{n \in \omega}, d_{\mathcal{Y}})$ as follows. First $d_{\mathcal{Y}} \upharpoonright \{\alpha_n\}_{n \in \omega} = d_{\mathcal{X}}$. The value of $d_{\mathcal{Y}}$ on the remaining points will be defined in stages, adding one point at a time. We will use finite injury with the following requirements:

$\mathcal{R}_e : W_e$ is not a name for a continuous bijection from $\mathcal{Y}$ to $\mathcal{X}$

The strategy for $\mathcal{R}_0$ works as follows: we look for an isolated point α_i in $\mathcal{X}$ and a radius q that witnesses that α_i is isolated. Note that the set of pairs α_i, q such that $B_i^q = \{\alpha_i\}$ is co-c.e, so our initial choice of α_i and q may need to be changed, but eventually we will have settled on an appropriate choice.

Once we have a candidate pair, (α_i, q) , we then wait until some stage s where we see $(B_j^r, B_k^p) \in W_{e,s}$ for some j, k, r, p with $B_k^p \subseteq^{\neq} B_i^q$. If $d_{\mathcal{Y}}$ has been defined on γ_j already, then we pick the next β_n on which $d_{\mathcal{Y}}$ has not yet been defined and define $d_{\mathcal{Y}}(\beta_n, x) = d_{\mathcal{Y}}(\gamma_j, x) + r/2$ for any $x \neq \beta_n$. In this way, we will have ensured that B_j^r contains at least 2 points, and thus W_0 is not a name of a bijection. If $d_{\mathcal{Y}}$ has not been defined on γ_j yet, then we can define $d_{\mathcal{Y}}(\gamma_j, x) = d_{\mathcal{Y}}(\alpha_i, x) + r/2$ for any $x \neq \gamma_j$ and achieve the same result.

Lower priority requirements will act in much the same way, but there are two restrictions put on them:

- First, we ask that when $\mathcal{R}_e$ defines $d_{\mathcal{Y}}$ on some point β_n it places β_n within 2^{-e} of the point α_i used to diagonalize.
- Second, we ask lower priority requirements to not add new points next to any points used by higher priority requirements. This is to ensure that each α_i only has finitely many points ‘attached’ to it. To do this, lower priority requirements may need to pick a new pair for (α_i, q) if γ_j was used by a higher priority requirement.

This completes the construction of $\mathcal{Y}$. It is hopefully clear that there is no computable continuous bijection from $\mathcal{Y}$ to $\mathcal{X}$ as presented. What may not be clear is that these spaces are in fact homeomorphic. For that, we appeal to Lemma 3.5 in the subsection below.

3.2. Doubling points in countable Polish spaces. Consider some countable Polish space $\mathcal{X} = (\{x_n\}_{n \in \omega}, d)$ (here $\{x_n\}_{n \in \omega}$ lists *all* the points in $\mathcal{X}$, not just a special set) and some function $f : \omega \rightarrow \omega$. We define a new extension space $\mathcal{Y} = \text{double}(\mathcal{X}, f)$ as follows: we add points $\{y_{n,m} : m < f(n)\}$ and define for each $y_{n,m}$, $d(y_{n,m}, z) = d(x_n, z) + 2^{-n}$.

It follows from the construction that the space $\mathcal{Y}$ is homeomorphic to $\text{double}(\mathcal{X}, f)$ for some f and ordering of points of $\mathcal{X}$.

Lemma 3.5. *For any countable Polish space $\mathcal{X} = (\{x_n\}_{n \in \omega}, d)$ and function $f : \omega \rightarrow \omega$, $\text{double}(\mathcal{X}, f) \cong \mathcal{X}$.*

Proof. We will build a bijection h from $\mathcal{Y} = \text{double}(\mathcal{X}, f)$ to $\mathcal{X}$. We will have h fix all limit points of $\mathcal{Y}$. Observe that the sequence $a_n \rightarrow a$ with each $a_n \in \mathcal{Y} \setminus (\mathcal{X} \cup \{a_k : k < n\})$ if and only if the sequence $b_n \rightarrow a$ where b_n is the closest point in $\mathcal{X}$ to a_n . This is because, for each k , there are

only finitely many points that has distance $> 2^{-k}$ from $\mathcal{X}$. So $\mathcal{Y}$ has no new limit points that are not already in $\mathcal{X}$.

We start by giving some example bijections that work in specific cases, then we explain how we can build a bijection for more complex spaces using transfinite recursion.

The first example is if $\mathcal{X}$ contains no limit points. In this case we have that both $\mathcal{X}$ and $\mathcal{Y}$ have the discrete topology, so any bijection will be a homeomorphism.

Now suppose that $\mathcal{X}$ contains exactly 1 limit point, x . We can partition the points in $\mathcal{X}$ into sets X_n defined by how close the points are to x . Define $X_n = \{x_k : d(x, x_k) \in [2^{-n}, 2^{-n+1})\}$ for $n > 0$ and $X_0 = \{x_k : d(x, x_k) \in [1, \infty)\}$.

If for every n , X_n is finite, then every distinct sequence of points (in $\mathcal{X}$ or in $\mathcal{Y}$) converges to x . So any bijection on the isolated points will work in this case.

If there are only finitely many n for which X_n is infinite, let n be the largest. Then we can take a bijection between the isolated points of $\mathcal{Y}$ that are attached to some point in some X_m for $m > n$, and the points in X_m , for $m > n$, as any sequence of these points converges to x . This bijection can be extended to the remaining points in any way, as no sequence in the left over points converges to x .

If there are infinitely many infinite X_n , then we can construct a homeomorphism by taking the union of a series of bijections, each restricted to some interval $X_n \cup \dots \cup X_{n+k}$ and the attached points in $\mathcal{Y}$.

We have argued how to build a bijection in the case where there is only one limit point. For the general proof we will partition the isolated points of $\mathcal{X}$ into countably many infinite sections, each with exactly 1 limit point. By building h on each of these sections like we have above, we can ensure that any sequence of points in one of these sections converges if and only if it converges after applying h .

The tricky part is what happens when considering sequences that belong to infinitely many distinct sections of our partition. In this case we compare our sequences to the sequence of limit points of corresponding sections. This will work, provided that for each k there are only finitely many sections with a diameter greater than 2^{-k} .

The construction of this partition will take α many stages for some countable ordinal α . At each stage s , we will be given a collection of isolated points U_s and will construct a section S_s from these remaining points. We keep a collection of isolated points U_s at each stage s . We start with U_0 as the set of all isolated points in $\mathcal{X}$. We define $U_{s+1} = U_s \setminus S_s$ and for limit stages, γ , we define $U_\gamma = \bigcap_{\beta < \gamma} U_\beta$.

At stage s , we define S_s as follows: we consider the boundary ∂U_s and ask what is the least n such that $x_n \in \partial U_s$ and x_n is isolated in ∂U_s . We then pick k such that $k > n$ and $B(x_n, 2^{-k}) \cap \partial U_s = \{x_n\}$. We then define section S_s of our partition to be $B(x_n, 2^{-k}) \cap U_s$.

If there is no such x_n then it must be that $\partial U_s = \emptyset$, so our construction can come to an end. We assign all remaining points in U_s to S_0 so that all isolated points remain in some section.

This completes the construction of our partition. For each section we can define a bijection as we did in the case of 1 limit point. We ensured that for each k there are only finitely many sections of diameter $> 2^{-k}$, so we will have that any sequence that picks points from distinct sections converges if and only if the sequence of limit points converges. So, h, h^{-1} constructed in this way will have the desired property of taking convergent sequences to convergent sequences, and hence extend to a homeomorphism. $\square$

4. THE BAIRE SPACE

We now consider $\mathcal{X} = \omega^\omega$ to be the Baire space. $\omega^{<\omega}$ shall be used to denote the standard Polish presentation of ω^ω . Each basic open ball of $\omega^{<\omega}$ is identified with a finite string σ , and we write $[\sigma] = \{x \in \omega^\omega \mid x \supset \sigma\}$. If $\sigma \neq \tau \in \omega^{<\omega}$ we let $\sigma \cap \tau$ be $\sigma \upharpoonright k$ where k is the least number such that $\sigma(k) \neq \tau(k)$. The standard metric on ω^ω is defined as $d_0(x, y) = 2^{-|x \cap y|}$, and is 0 if $x = y$. For each $\sigma * k \in \omega^{<\omega}$ let $a_{\sigma * k} = \sigma * k * 1^\omega$ if $k = 0$, otherwise let $a_{\sigma * k} = \sigma * k * 0^\omega$. Then, the standard presentation $\omega^{<\omega}$ has $\{a_\sigma \mid \sigma \in \omega^\omega, |\sigma| > 0\}$ as its set of special points and is equipped with the ultrametric d_0 .

Before analysing the upper and lower bounds on the categoricity of ω^ω , we first turn our attention to the much simpler Cantor space. Using [HTMN20, Lemma 4.2], or [HKS23, Lemma 2.1], it is easy to see that between any two computable Polish presentations of Cantor space, $\mathbf{0}'$ suffices to compute a homeomorphism between them. The key idea is in producing increasingly fine covers using clopen sets of the given presentation of Cantor space. Then using a standard back-and-forth argument, one may then construct a computable homeomorphism between any two given presentations with ‘nice’ covers as given by $\mathbf{0}'$. Conversely, since the standard presentation $2^{<\omega}$ of Cantor space is computably compact, but general computable presentations of Cantor space are merely $\emptyset'$ -computably compact, applying Lemma 2.5, one may obtain that the degree of categoricity of Cantor space is $\mathbf{0}'$.

A natural approach to analysing the upper and lower bounds on the categoricity of ω^ω would be to generalise the ideas used in the Cantor space. Whilst ω^ω is non-compact, meaning that clopen subsets of ω^ω may not be a union of finitely many open balls in a given Polish presentation of ω^ω , one might suspect that this does not pose any issues when attempting to build a computable homeomorphism between ω^ω and an arbitrary computable Polish presentation $\mathcal{M}$. A naive attempt at building a computable homeomorphism $f : \omega^\omega \rightarrow \mathcal{M}$ might proceed as follows. Assume that we have identified open balls $B_0, \dots, B_k$ such that $f([i]) = B_i$ for $i \leq k$ where $a \notin \bigcup_{i \leq k} B_i$ for some special point a . We find some open ball B_{k+1} containing the next special point of $\mathcal{M}$, B_{k+1} is disjoint from $\bigcup_{i \leq k} B_i$ and $\bigcup_{i \leq k+1} B_i$ is again non-empty (this can be done as $\mathcal{M}$ is perfect). We then set $f([k+1]) = B_{k+1}$. Repeat this process inside each basic open ball.

Unlike the situation in the Cantor space, we are never forced to commit to a finite open cover of $\mathcal{M}$ when building a homeomorphism f . We can map each countable cover of ω^ω to a sequence of mutually disjoint open balls which can be found effectively (like described above). Furthermore, $\bigcup_{i \in \omega} B_i$ will contain all the special points of $\mathcal{M}$, so what could prevent f from being a homeomorphism?

The issue with the naive approach above is that when attempting to cover all special points of $\mathcal{M}$ we might end up with $\liminf_i r(B_i) = 0$. So even though $\bigcup_{i \in \omega} B_i$ might contain all special points of $\mathcal{M}$, it might not cover $\overline{\mathcal{M}}$. For instance, if the first open ball $f([0]) = B_0$ that we pick is not even closed, then there is no chance for us to extend f to a homeomorphism. An arbitrary computable presentation $\mathcal{M}$ of ω^ω could be badly behaved in the sense that there is no effective open basis of clopen sets. In order to make the analysis of ω^ω more tractable, we restrict our study to the presentations of ω^ω that are well-behaved:

Definition 4.1. Let $\mathcal{M}$ be a computable Polish presentation of ω^ω . $\mathcal{M}$ is called *effectively zero-dimensional* if there is a c.e. set $\mathcal{V}$ such that

- For every $\langle e, k \rangle \in \mathcal{V}$, we have $\mathcal{O}(W_e)$ and $\mathcal{O}(W_k)$ is a partition of $\overline{\mathcal{M}}$.
- For every $\langle e, k \rangle \in \mathcal{V}$, we have $\text{diam}(\mathcal{O}(W_e))$ is a computable real, uniformly in $\langle e, k \rangle$.

- For every $x \in \overline{\mathcal{M}}$ and every rational number $\varepsilon > 0$, there is some $\langle e, k \rangle \in \mathcal{V}$ such that $x \in \mathcal{O}(W_e)$ and $\text{diam}(\mathcal{O}(W_e)) < \varepsilon$.

That is, $\mathcal{M}$ is effectively zero-dimensional if it has a c.e. basis of clopen sets where each clopen set in the basis is both effectively open and effectively closed. Furthermore, the second and third conditions ensure that given the name of any point x and the name of any open ball B containing x , we can effectively find the index of a clopen set U from $\mathcal{V}$ such that $x \in U \subseteq^\mathfrak{f} B$. As usual we can relativize this notion to any oracle X .

We say that $\mathcal{M}$ is *strongly effectively zero-dimensional* if it is effectively zero-dimensional and additionally, for each $\langle e, k \rangle \in \mathcal{V}$, W_e is finite and we are given the canonical index for W_e .

Example 4.2. Any computable Polish presentation $\mathcal{M}$ of ω^ω where every basic open ball of $\mathcal{M}$ is (uniformly) effectively closed in $\mathcal{M}$ is a strongly effectively zero-dimensional presentation. For instance, the standard Polish presentation $\omega^{<\omega}$ is strongly effectively zero-dimensional.

Example 4.3. Any computable Polish structure with a rational-valued ultrametric is strongly effectively zero-dimensional; in such a space, every basic open ball is effectively closed. For instance, the standard presentation $\omega^{<\omega}$. To see this, note that in an ultrametric space, intersecting balls are contained in each other. Thus, given any two basic balls B_p^i and B_q^j , $B_p^i \cap B_q^j = \emptyset$ if and only if $d(\alpha_i, \alpha_j) \geq \max\{p, q\}$; the latter is recursive since d is rational-valued.

Example 4.4. Every computable Polish presentation $\mathcal{M}$ of the Cantor space 2^ω is strongly $\emptyset'$ -effectively zero-dimensional. To see this, we enumerate $\langle \langle B_0, \dots, B_N \rangle, \langle C_0, \dots, C_M \rangle \rangle$ into $\mathcal{V}$ iff $(\bigcup_{i \leq N} B_i) \cap (\bigcup_{i \leq M} C_i) = \emptyset$, and $(\bigcup_{i \leq N} B_i) \cup (\bigcup_{i \leq M} C_i) = \overline{\mathcal{M}}$. Obviously, $\mathcal{V}$ is c.e. relative to $\emptyset'$.

Now given any $x \in \overline{\mathcal{M}}$ and rational $\varepsilon > 0$, we fix a clopen set $D \subset \overline{\mathcal{M}}$ such that $x \in D \subseteq B(x, \frac{\varepsilon}{3})$ (D exists since $\overline{\mathcal{M}}$ is zero-dimensional). Since $D \subseteq B(x, \frac{\varepsilon}{3})$, we can cover D by basic open balls that are formally included in $B(x, \frac{\varepsilon}{3})$, and hence we can write $D = \bigcup_{i \leq N} B_i$ for some $\langle B_0, \dots, B_N \rangle \in \mathcal{V}$ where $\bigcup_{i \leq N} B_i \subseteq^\mathfrak{f} B(x, \frac{\varepsilon}{3})$. This formal inclusion implies that the diameter of the collection $\langle B_0, \dots, B_N \rangle$ is less than ε .

The first thing to check is that Definition 4.1 is not too strong to be a trivial notion. For instance, if the (strongly) effectively zero-dimensional presentations of ω^ω are unique up to computable homeomorphism, then Definition 4.1 will not be helpful in analyzing the categoricity of ω^ω .

We first test this for the Cantor space. It turns out that the Cantor space, when restricted to strongly effectively zero-dimensional presentations, still has degree of categoricity $\mathbf{0}'$:

Proposition 4.5. *The Cantor space, when restricted to strongly effectively zero-dimensional presentations, has degree of categoricity $\mathbf{0}'$.*

Proof. As discussed earlier, any two computable presentations of 2^ω are $\mathbf{0}'$ -homeomorphic, so we only need to show that $\mathbf{0}'$ is a lowerbound for strongly effectively zero-dimensional spaces. Define the tree $T \subseteq 3^{<\omega}$ to be the set of all $\sigma \in 3^{<\omega}$ such that for every $i < |\sigma|$ where $\sigma(i) = 3$, we have $i \in \emptyset'$.

Since T is a c.e. tree, let $\mathcal{M}$ be identified with a computable set representing $\{a_\sigma \mid \sigma \in T, |\sigma| > 0\}$. Hence $\overline{\mathcal{M}}$ (as a subspace of ω^ω) has a computable Polish presentation. If $\sigma \in T$ then $\sigma * 0$ and $\sigma * 1$ are both in T and so T has no dead ends, which means that $\overline{\mathcal{M}}$ is perfect and hence $\overline{\mathcal{M}} \cong [T]$. However, it is also clear that $[T] \cong 2^\omega$, since T is uniformly 2-branching or 3-branching at each level. Furthermore, as T is c.e., each basic open ball is effectively closed, uniformly in an index for the open ball. So, $\mathcal{M}$ is a strongly effectively zero-dimensional presentation of 2^ω .

Now suppose that some homeomorphism $f : 2^\omega \rightarrow \overline{\mathcal{M}}$ is D -computable. Now given n we search for a stage s where $\bigcup \{f^{-1}(B(a_\sigma, 2^{-n})) [s] \mid \sigma \in T[s], |\sigma| = n+1\} \supseteq 2^\omega$. We can find such a stage s effectively in D , as 2^ω is computably compact. Now if n enters $\emptyset'$ after stage s then there is some $\sigma \in T$ that is incomparable with every string in $T[s]$, which means that $\bigcup \{B(a_\sigma, 2^{-n}) \mid \sigma \in T[s], |\sigma| = n+1\} \neq \overline{\mathcal{M}}$, a contradiction. So, $\emptyset' \leq_T D$. $\square$

We next show that not every computable Polish presentation of the Baire space (and the Cantor space) is strongly effectively zero-dimensional:

Proposition 4.6. *There is a computable Polish presentation of ω^ω (and also of 2^ω) that is not strongly effectively zero-dimensional.*

Proof. We construct a computable Polish presentation $\mathcal{M}$ for 2^ω that is not strongly effectively zero-dimensional; the construction can be easily modified to give a presentation for ω^ω . The structure of $\mathcal{M}$ will be organised in the form of a subtree T of $3^{<\omega}$, with the dense subset $\{a_\sigma \mid \sigma \in T, |\sigma| > 0\}$. However, we will need to modify the metric d from d_0 so that it is not an ultrametric (see Example 4.3).

For each e , let W_e be the e -th c.e. set. We let $Q_e \subseteq 3^{<\omega}$ be determined in the following way. Search for the first $\langle B(a_{\sigma_0}, r_0), \dots, B(a_{\sigma_j}, r_j) \rangle \in W_e$ such that for every $i \leq j$ we have $r_i < 2^{-e-2}$, $\sigma_i \supseteq 0^e * 1$ and $\sigma_i \in 2^{<\omega}$. (To do the diagonalisation we shall not need to use the fact that the complement of $B(a_{\sigma_0}, r_0) \cup \dots \cup B(a_{\sigma_j}, r_j)$ is also given effectively).

Pick i_0 such that r_{i_0} is the largest (if there is more than one choice, pick any one). Let $\tau = a_{\sigma_{i_0}} \upharpoonright M$, where M is the least such that $2^{-M} \leq r_{i_0}$. If this can be found, we let $Q_e = \{\tau * 2 * \sigma \mid \sigma \in 2^{<\omega}\}$, otherwise let $Q_e = \emptyset$. Let $T = 2^{<\omega} \cup (\bigcup_{e \in \omega} Q_e)$. Then T is a subtree of $3^{<\omega}$, and note that Q_e s are mutually disjoint since $M \geq e+2$. Thus the different τ s found by different Q_e s are mutually incomparable.

Now the structure $\mathcal{M}$ will have the set $\{a_\sigma \mid \sigma \in T\}$ as its set of special points. Recall that the standard metric on ω^ω is defined as $d_0(x, y) = 2^{-|x \cap y|}$, and is 0 if $x = y$. We define the metric d for $\mathcal{M}$ as follows:

- If $\sigma = \sigma'$ then $d(a_\sigma, a_{\sigma'}) = 0$.
- If there is some e (which will be unique) such that $Q_e \neq \emptyset$ (so that the corresponding M , τ , r_{i_0} and σ_{i_0} exist) and $\sigma \supseteq \tau * 2$ and $a_{\sigma'} \supset \tau * k$ for $k = 0$ or 1 , then we define $d(a_\sigma, a_{\sigma'}) = d(a_{\sigma'}, a_\sigma) = r_{i_0} - d_0(a_\sigma, a_{\tau * 2})$.
- Otherwise, define $d(a_\sigma, a_{\sigma'}) = d_0(a_\sigma, a_{\sigma'})$.

We verify that d is a metric on the set $\{a_\sigma \mid \sigma \in T\}$. For any three points a_σ, a_η and a_ρ , we will have the ultrametric property $d(a_\sigma, a_\eta) \leq \max\{d(a_\sigma, a_\rho), d(a_\rho, a_\eta)\}$ except in the following three situations:

- There is some e such that $Q_e \neq \emptyset$ and $a_\sigma, a_\rho \supseteq \tau * 2$ and $a_\eta \supseteq \tau * 0$ in the Q_e -cone (the case $a_\eta \supseteq \tau * 1$ is identical). In this case, $d(a_\sigma, a_\rho) = d_0(a_\sigma, a_\rho)$, $d(a_\sigma, a_\eta) = r_{i_0} - d_0(a_\sigma, a_{\tau * 2})$ and $d(a_\rho, a_\eta) = r_{i_0} - d_0(a_\rho, a_{\tau * 2})$. Hence it follows that $d(a_\sigma, a_\eta) \leq d(a_\sigma, a_\rho) + d(a_\rho, a_\eta)$ and $d(a_\rho, a_\eta) \leq d(a_\sigma, a_\eta) + d(a_\sigma, a_\rho)$ by writing in terms of d_0 . By the choice of r_{i_0} we have $2^{-M} \leq r_{i_0}$ and so we also have $d(a_\sigma, a_\rho) \leq d(a_\sigma, a_\eta) + d(a_\rho, a_\eta)$.
- There is some e such that $Q_e \neq \emptyset$ and $a_\sigma \supseteq \tau * 2$, $a_\eta \supseteq \tau * 0$ and a_ρ does not extend τ . In this case, $d(a_\sigma, a_\rho) = d(a_\rho, a_\eta) \geq 2^{-M+1}$, and $d(a_\sigma, a_\eta) = r_{i_0} - d_0(a_\sigma, a_{\tau * 2}) \leq r_{i_0} < 2^{-M+1} \leq \max\{d(a_\sigma, a_\rho), d(a_\rho, a_\eta)\}$. (This case also has the ultrametric property).

- There is some e such that $Q_e \neq \emptyset$ and $a_\sigma \supseteq \tau * 2$, $a_\eta \supseteq \tau * 0$ and $a_\rho \supseteq \tau * 1$. In this case, $d(a_\rho, a_\eta) = 2^{-M}$, and $d(a_\sigma, a_\rho) = d(a_\sigma, a_\eta) = r_{i_0} - d_0(a_\sigma, a_{\tau*2}) \geq 2^{-M} - 2^{-M-1} = 2^{-M-1}$. Thus, $d(a_\rho, a_\eta) \leq d(a_\sigma, a_\rho) + d(a_\sigma, a_\eta)$.

Since T is a c.e. tree and d is a metric on T , $\mathcal{M}$ is a computable Polish structure. Observe that if $k = |a_\eta \cap a_\rho|$, then $d(a_\eta, a_\rho)$ is either $d_0(a_\eta, a_\rho)$ or $r_{i_0} - d_0(a_\eta, a_{\tau*2})$ for some parameters M , τ , r_{i_0} and σ_{i_0} . In the first case $d_0(a_\eta, a_\rho) = 2^{-k}$, and in the second case we must have $M = k$ and $2^{-M+1} > r_{i_0} \geq r_{i_0} - d_0(a_\eta, a_{\tau*2}) \geq 2^{-M} - 2^{-M-1} = 2^{-M-1}$, and thus $2^{-k-1} \leq d(a_\eta, a_\rho) < 2^{-k+1}$ holds in both cases. Thus our modification to the standard metric on ω^ω does not cause “new limit points” to be formed.

More specifically, we claim that $f : x \in [T] \mapsto \lim_n a_{x \upharpoonright n}$ is a homeomorphism. As $\sigma \in T$ implies that $\sigma * 0, \sigma * 1 \in T$, thus every special point is also a limit point, and so f is surjective. Now consider $S = \{\sigma \in T \mid \sigma \neq \tau * k \text{ and } \sigma \neq \tau \text{ for any } \tau \text{ found by a } Q_e \text{ and } k = 0, 1, 2\}$. We claim that if $\sigma \in S$, then $\overline{\{a_\eta \mid \eta \supseteq \sigma\}} = B(a_\sigma, 2^{-|\sigma|+1})$. For any $\eta \supseteq \sigma$, if $a_\eta \cap a_\sigma = \sigma$ then $d(a_\eta, a_\sigma) = d_0(a_\eta, a_\sigma) = 2^{-|\sigma|}$ (since $\sigma \neq \tau$), and if $a_\eta \cap a_\sigma \neq \sigma$ then $d(a_\eta, a_\sigma) < 2^{-|a_\eta \cap a_\sigma|+1} \leq 2^{-|\sigma|}$. Therefore, $\overline{\{a_\eta \mid \eta \supseteq \sigma\}} \subseteq B(a_\sigma, 2^{-|\sigma|+1})$. Conversely suppose that $\eta \mid \sigma$. If $|\eta \cap \sigma| + 1 = |\sigma|$ then as $\sigma \neq \tau * k$ for any $k = 0, 1, 2$, we conclude that $d(a_\eta, a_\sigma) = d_0(a_\eta, a_\sigma) = 2^{-|\eta \cap \sigma|} = 2^{-|\sigma|+1}$. Otherwise, $|\eta \cap \sigma| + 1 < |\sigma|$ and $d(a_\eta, a_\sigma) \geq 2^{-|a_\eta \cap a_\sigma|+1} = 2^{-|\eta \cap \sigma|+1} \geq 2^{-|\sigma|+1}$. Therefore, if $\lim_n a_{x \upharpoonright n} \notin \overline{\{a_\eta \mid \eta \supseteq \sigma\}}$ then $d(\lim_n a_{x \upharpoonright n}, a_\sigma) \geq 2^{-|\sigma|+1}$.

Notice that the collection $\{\overline{\{a_\eta \mid \eta \supseteq \sigma\}} \mid \sigma \in S\}$ forms a co-c.e. basis of clopen sets for $\overline{\mathcal{M}}$. Now for any $x \in [T]$, $x \upharpoonright k \in S$ for all except at most two k . Since $f([\sigma]) = \overline{\{a_\eta \mid \eta \supseteq \sigma\}}$ for any $\sigma \in T$, f is continuous and open.

We now argue that $\mathcal{M}$ is not strongly effectively zero-dimensional. If it were then there would be a c.e. set W_e satisfying Definition 4.1. Taking $x = a_{0^e * 1}$ and $\varepsilon = 2^{-e-3}$ there must be some $\langle B(a_{\sigma_0}, r_0), \dots, B(a_{\sigma_j}, r_j) \rangle \in W_e$ satisfying Definition 4.1. Hence, $Q_e \neq \emptyset$. By the maximality of r_{i_0} and the fact that $\sigma_0, \dots, \sigma_j \in 2^{<\omega}$, we see that $a_{\tau*2} \notin B(a_{\sigma_0}, r_0) \cup \dots \cup B(a_{\sigma_j}, r_j)$. However the sequence $\langle a_{\tau*2*0^k} \rangle_{k \in \omega}$ is a sequence of points in $B(a_{\sigma_{i_0}}, r_{i_0})$ with limit $a_{\tau*2}$. Hence, $B(a_{\sigma_0}, r_0) \cup \dots \cup B(a_{\sigma_j}, r_j)$ is not closed, much less effectively closed, a contradiction. $\square$

Remark 4.7. Although we prove Proposition 4.6 only for strongly effectively zero-dimensional presentations of the Cantor space and the Baire space, we conjecture that Proposition 4.6 also holds for effectively zero-dimensional presentations. We now sketch a strategy which may be used to build a computable presentation of 2^ω that is not effectively zero-dimensional. (Again, the strategy can be easily modified for ω^ω).

As in the proof of Proposition 4.6, we carry out each diagonalisation strategy in a different cone, so the different strategies work independently. In the cone above $0^d * 1$ we search for the first $\langle e, k \rangle \in W_d$ such that $\mathcal{O}(W_e)$ contains some ball $B_0 \subseteq [0^d * 1]$ and $\mathcal{O}(W_k)$ contains some ball $\hat{B}_0 \subseteq [0^d * 1]$. If B_0 and $\hat{B}_0$ are not formally disjoint then we can introduce a new copy C of ω^ω and declare the metric on $\mathcal{M}$ so that C is formally contained inside $B_0 \cap \hat{B}_0$; this kills W_e as $\mathcal{O}(W_e)$ and $\mathcal{O}(W_k)$ are not disjoint. If B_0 and $\hat{B}_0$ are formally disjoint we introduce a new copy C of ω^ω and declare the metric on $\mathcal{M}$ so that $d(x, y) = r(B_0) + \frac{d(b_0, \hat{b}_0) - r(B_0) - r(\hat{B}_0)}{2}$ for every $x \in C$ and $y \in B_0$ and $d(x, y) = r(\hat{B}_0) + \frac{d(b_0, \hat{b}_0) - r(B_0) - r(\hat{B}_0)}{2}$ for every $x \in C$ and $y \in \hat{B}_0$. Here, b_0 and $\hat{b}_0$ are the centers of B_0 and $\hat{B}_0$ respectively. That is, we put C in the center of the “gap” between B_0 and $\hat{B}_0$.

Next we wait for the first ball A appearing in either $\mathcal{O}(W_e)$ or $\mathcal{O}(W_k)$ to be contained in C . If A is in $\mathcal{O}(W_e)$ and is not formally disjoint from $\hat{B}_0$ we can diagonalise as above. Otherwise we will introduce a new copy C' of ω^ω and put it in the gap between A and $\hat{B}_0$. In this case we label $A = B_1$. Otherwise if A is in $\mathcal{O}(W_k)$ we can also either diagonalise or squeeze a new copy C' in the gap between A and B_0 , and we label $A = \hat{B}_1$. Continuing this way, the strategy will have one of the following outcomes:

- After placing finitely many new disconnected components $B_0, \dots, B_i$ and $\hat{B}_0, \dots, \hat{B}_j$ we diagonalise as above. Then we may identify $B_0, \dots, B_i, \hat{B}_0, \dots, \hat{B}_j$ with a finite cover of $[0^d * 1 * 0]$ and so $\overline{\mathcal{M}} \cong 2^\omega$.
- We add finitely many $B_0, \dots, B_i$ but infinitely many $\hat{B}_0, \dots$. Then $\overline{\mathcal{M}}$ contains a limit point x of $\bigcup_{i \in \omega} \hat{B}_i$, but $x \notin \mathcal{O}(W_e) \cup \mathcal{O}(W_k)$. To show $\overline{\mathcal{M}} \cong 2^\omega$ we identify $\hat{B}_i$ with $[0^d * 1 * 0^{i+1} 1]$ for each $i \in \omega$, and x with $0^d * 1 * 0^\omega$.
- We add finitely many $\hat{B}_0, \dots, \hat{B}_i$ but infinitely many $B_0, \dots$; this case is symmetric to the one above.
- We add infinitely many $B_0, \dots$ and infinitely many $\hat{B}_0, \dots$. Then $\overline{\mathcal{M}}$ contains a limit point x of $\bigcup_{i \in \omega} B_i \cup \hat{B}_i$, but $x \notin \mathcal{O}(W_e) \cup \mathcal{O}(W_k)$. To show $\overline{\mathcal{M}} \cong 2^\omega$ we identify B_i with $[0^d * 1 * 0^{2i+2} 1]$ and $\hat{B}_i$ with $[0^d * 1 * 0^{2i+1} 1]$ for each $i \in \omega$, and x with $0^d * 1 * 0^\omega$.

We leave it to the reader to verify that this strategy works.

Even though there are fewer (strongly) effectively zero-dimensional presentations compared to all effective presentations, Proposition 4.5 shows that restricting to strongly effectively zero-dimensional presentations for the Cantor space does not affect the degree of categoricity. In the next theorem (Theorem 4.8), we will show that restricting to strongly zero-dimensional presentations for the Baire space is also not unreasonable.

Recall the discussion at the beginning of Section 4 where we gave a naive (and incorrect) strategy to show that the Baire space is computably categorical. In the next theorem (Theorem 4.8) we show that no such strategy can succeed; in fact we show that the Baire space is not even $\mathbf{0}'$ -categorical. Furthermore, this can be witnessed by a pair of strongly effectively zero-dimensional presentations for ω^ω . This confirms that considering only effectively zero-dimensional presentations for the Baire space is not overly restrictive:

Theorem 4.8. *The Baire space ω^ω is not $\mathbf{0}'$ -categorical. In fact, there is a strongly effectively zero-dimensional computable presentation $\overline{\mathcal{M}}$ such that $\overline{\mathcal{M}}$ and ω^ω are not $\mathbf{0}'$ -homeomorphic.*

Proof. We will build a c.e. subtree $T \subseteq \omega^{<\omega}$. More specifically, we will be providing an effective enumeration $\langle T_s \rangle$ for T , and let a_k be associated with the k -th element enumerated into T for each $k \in \omega$. We define $d(a_k, a_l) = d_0(a_\sigma, a_\rho)$ where a_k and a_l are associated with a_σ and a_ρ respectively. At the end we take $\mathcal{M}$ to have $\{a_0, a_1, \dots\}$ as the set of special points with metric d . The basic ball B_p^i is taken with respect to this presentation; if $\sigma \in T$ we will also write B_p^σ in place of B_p^i , where σ is associated with a_i . Since we are using the standard metric for $\overline{\mathcal{M}}$, we may assume without loss of generality that all basic balls have radius $p = 2^{-r}$ for some r . We shall be comparing $\overline{\mathcal{M}}$ to the standard presentation ω^ω . To prevent confusion, we will write $[\sigma]$ to stand for the basic ball $B(a_\sigma, 2^{-|\sigma|+1})$ of the standard presentation ω^ω .

Let $\langle W_e, V_e \rangle$ be the e -th pair of Σ_2^0 sets. We effectively approximate a Σ_2^0 set W by $W(x) = \liminf_s W_s(x)$. To diagonalize against $\langle W_e, V_e \rangle$ we will work inside $B_{2^{-e}}^{0_e}$ of $\overline{\mathcal{M}}$; the purpose of restricting the influence of the diagonalization procedure for $\langle W_e, V_e \rangle$ in this way is to ensure that

$[T]$ is homeomorphic to ω^ω . The pair $\langle W_e, V_e \rangle$ represents some potential $\mathbf{O}'$ -effective homeomorphism f_e . As such, we view W_e and V_e as consisting of elements of the form $\langle B_p^\sigma, [\tau] \rangle$, where $f_e(B_p^\sigma) = \bigcup \{[\tau] \mid \langle B_p^\sigma, [\tau] \rangle \in W_e\}$ and $f_e^{-1}([\tau]) = \bigcup \{B_p^\sigma \mid \langle B_p^\sigma, [\tau] \rangle \in V_e\}$. We write $\Gamma_{e,s}$ to be the set of all strings η of length $< s$ such that there is some $\langle B_{2^{-e}}^{0^e}, [\tau] \rangle \in W_{e,s}$ and some $\langle B_{2^{-r}}^\sigma, [\tau * k] \rangle \in V_{e,s}$ where $\eta \supseteq a_\sigma \upharpoonright r+1$ and $\eta \in T_s$. We call $\langle B_{2^{-e}}^{0^e}, [\tau] \rangle$ and $\langle B_{2^{-r}}^\sigma, [\tau * k] \rangle$ a witness for $\eta \in \Gamma_{e,s}$. We say that η is stably in Γ_e from stage t to stage s if $\eta \in \Gamma_{e,v}$ for every $t \leq v \leq s$ with the same witness, and say that η is stably in Γ_e if it is forever in Γ_e with the same witness after some stage. This definition is obviously used to detect membership in a Σ_2^0 set. The usual conventions apply; approximations at stage s only look at (strings encoded by) numbers less than s .

For each e, s and $\sigma \supseteq 0^e * 1$ with $\sigma \in T_s$, we say that s is $\langle e, \sigma \rangle$ -*expansionary* if one of the following holds:

- (i) s is equal to the code number of σ .
- (ii) There is some $\eta \subseteq \sigma$ such that $\eta \in \Gamma_e$ stably from $|\sigma|$ to s .
- (iii) There is a largest $\langle e, \sigma \rangle$ -expansionary stage $t < s$ and some string $\eta \supseteq 0^e * 1$ to the left of σ such that every η' comparable with η is not stably in Γ_e from t to s .

We let $T_0 = 2^{<\omega}$. At stage $s+1$ we do the following for each $\sigma \in T_s$ such that $|\sigma| < s$ and there is some e such that $0^e * 1 \subseteq \sigma$: If $s+1$ is $\langle e, \sigma \rangle$ -expansionary, we put $\sigma * k * \rho$ into T_{s+1} where k is the least such that $\sigma * k \notin T_s$ and $\rho \in 2^{<\omega}$.

Clearly $\mathcal{M}$ is computable as T is c.e.. As usual the map $x \in [T] \mapsto \lim_n a_{x \upharpoonright n}$ is a homeomorphism; the surjectivity of this map follows from the fact that $\sigma \in T \Rightarrow \sigma * 0, \sigma * 1 \in T$. Now we check that $[T] \cong \omega^\omega$.

Claim 4.9. *Let $e \in \omega$ and $\sigma \in T$ such that $\sigma \supseteq 0^e * 1$ be given. Then there is some $\tau \supseteq \sigma$ such that $\tau \in T$ and there are infinitely many $\langle e, \tau \rangle$ -expansionary stages.*

Proof. Fix $\sigma \in T$ and let e be such that $0^e * 1 \subseteq \sigma$. Consider the set $G = \{\sigma * \eta \mid \sigma * \eta \in T\}$. There are two cases. First, suppose that G contains some τ such that there is some $\tau' \subseteq \tau$ where τ' is stably in Γ_e after some stage t . Then there are infinitely many $\langle e, \tau * 0^t \rangle$ -expansionary stages. On the other hand if G contains no such τ then we claim that there are infinitely many $\langle e, \sigma * 1 \rangle$ -expansionary stages. Let s_1 be any $\langle e, \sigma * 1 \rangle$ -expansionary stage. No element comparable with $\sigma * 0$ can be stably in Γ_e by our case assumption. Since Γ_{e,s_1} is finite, item (iii) in the definition of a $\langle e, \sigma * 1 \rangle$ -expansionary stage will eventually be met after s_0 . $\square$

Claim 4.9 tells us that the set of nodes of T with infinitely many successors is dense in T . This allows us to build a homeomorphism $f : \omega^\omega \rightarrow [T]$ easily: Assume that $f([\sigma]) = [\tau]$ has been defined. Let τ' be a minimal string such that $\tau' \supseteq \tau$ and τ' has infinitely many successors in T . Then $\tau' \upharpoonright j$ has finitely many successors in T for every $|\tau| \leq j < |\tau'|$. We simply define $f([\sigma * 0]), f([\sigma * 1]), \dots, f([\sigma * N])$ to be the cylinders generated by the finite set of successors of $\tau' \upharpoonright j$ for $|\tau| \leq j < |\tau'|$ which are incomparable with τ' . Then we take $f([\sigma * (N + k + 1)]) = [\tau' * k]$ for all $k \in \omega$. If $\tau = \tau'$ then we define f in the obvious way.

$\overline{\mathcal{M}}$ is strongly effectively zero-dimensional as T is c.e.. Thus it only remains to verify that $\overline{\mathcal{M}}$ and ω^ω are not $\mathbf{O}'$ -homeomorphic. Suppose that they are. Then there is some e such that the pair $\langle W_e, V_e \rangle$ represents the homeomorphism $f_e : \overline{\mathcal{M}} \rightarrow \omega^\omega$. We now define a path $x \in [T]$ to help us get to a contradiction.

First let $x \upharpoonright e+1 = 0^e * 1$. We claim that there does not exist a stage s and some finite collection $\eta_0, \dots, \eta_k$ stably in $\Gamma_{e,s}$ such that $[T] \cap [x \upharpoonright e+1] \subseteq [\eta_0] \cup \dots \cup [\eta_k]$ (we call this a finite cover of $[T] \cap [x \upharpoonright e+1]$ by Γ_e). Suppose $s, \eta_0, \dots, \eta_k$ exist. Let τ_0 be any minimal element such that

$\langle B_{2-e}^{0^e}, [\tau_0] \rangle \in W_{e,s}$ stably, and since $V_{e,s}$ is finite, let k_0 be the largest such that $\langle B_{2-r}^\sigma, [\tau_0 * k_0] \rangle \in V_{e,s}$. Now since $[\tau_0] \subseteq f_e(B_{2-e}^{0^e})$ it follows that there is some $z \in [T] \cap [x \upharpoonright e+1]$ such that $f_e(\lim_n a_{z \upharpoonright n}) = \tau_0 * (k_0 + 1) * 0^\omega$. Since $[T] \cap [x \upharpoonright e+1] \subseteq [\eta_0] \cup \dots \cup [\eta_k]$, we let $\langle B_{2-e}^{0^e}, [\tau'] \rangle$ and $\langle B_{2-r}^\sigma, [\tau' * k'] \rangle$ be stable elements of W_e and V_e respectively such that $\lim_n a_{z \upharpoonright n} \in B_{2-r}^\sigma$. This gives us that $\tau_0 * (k_0 + 1) * 0^\omega \in [\tau' * k']$. By the minimality of τ_0 we must have $\tau_0 \subseteq \tau'$ or $\tau_0 \upharpoonright \tau'$. The latter is impossible, and the former contradicts the maximality of k_0 . Thus, no finite cover of $[T] \cap [x \upharpoonright e+1]$ by Γ_e is possible.

Next, we show that there are only finitely many $\langle e, x \upharpoonright e+1 \rangle$ -expansionary stages. Since every string lexicographically left of $x \upharpoonright e+1$ cannot extend $0^e * 1$, only item (ii) (in the definition of a $\langle e, x \upharpoonright e+1 \rangle$ -expansionary stage) can apply infinitely often. However, this is also impossible since there is no finite cover of $[T] \cap [x \upharpoonright e+1]$ by Γ_e .

Now inductively assume that $x \upharpoonright l$ has been defined for some $l > e$ with $x \upharpoonright l \in T$ and with the following two properties:

- There are only finitely many $\langle e, x \upharpoonright l \rangle$ -expansionary stages.
- There is no finite cover of $[T] \cap [x \upharpoonright l]$ by Γ_e .

We wish to now define $x(l)$. Since there are only finitely many $\langle e, x \upharpoonright l \rangle$ -expansionary stages, there are only finitely many successors of $x \upharpoonright l$ in T , and since there is no finite cover of $[T] \cap [x \upharpoonright l]$ by Γ_e , the same must be true of $[T] \cap [(x \upharpoonright l) * b]$ for some least b . Now we check that there can be only finitely many $\langle e, (x \upharpoonright l) * b \rangle$ -expansionary stages. Since there are only finitely many $\langle e, x \upharpoonright l \rangle$ -expansionary stages, and since b is minimal, item (iii) (in the definition of a $\langle e, (x \upharpoonright l) * b \rangle$ -expansionary stage) cannot hold infinitely often. Since there is no finite cover of $[T] \cap [(x \upharpoonright l) * b]$ by Γ_e , item (ii) cannot hold infinitely often. Thus we can take $x(l) = b$.

Now having obtained the path $x \in [T]$, we see that no initial segment of x can be stably in Γ_e , otherwise Γ_e would be able to finitely cover $[T] \cap [x \upharpoonright l]$ for some l . This contradicts $\langle W_e, V_e \rangle$ being a representation of f_e . Therefore, $\overline{\mathcal{M}}$ and ω^ω are not $\mathbf{0}'$ -homeomorphic. $\square$

Now that we know the Baire space is not computably categorical, we can turn to the interesting question as to what the degree of categoricity of the Baire space is. Computing the degree of categoricity of a structure typically consists of two parts; finding an upperbound and finding a lowerbound for categoricity of the structure.

We first consider the upperbound on categoricity of the Baire space. As mentioned at the beginning of Section 4, one obstacle towards constructing a computable homeomorphism between ω^ω and an arbitrary presentation $\overline{\mathcal{M}}$ is that it may be difficult to locate clopen sets in $\overline{\mathcal{M}}$ effectively. In the next theorem (Theorem 4.12) we show that with two jumps, this will be the only obstacle; if X can enumerate an effective clopen basis for $\overline{\mathcal{M}}$ then $\overline{\mathcal{M}}$ and ω^ω are X'' -homeomorphic. The alert reader will recognize this as a sort of effectivization of the Alexandrov-Urysohn Theorem. As a corollary, we conclude that any two effectively zero-dimensional presentations of the Baire space are $\mathbf{0}''$ -homeomorphic. By Theorem 4.8 this upperbound cannot be reduced to $\mathbf{0}'$.

We first require a lemma:

Lemma 4.10. *Let X and Y be oracles and $\overline{\mathcal{M}}$ and $\overline{\mathcal{N}}$ be X -computable Polish structures and $F : \overline{\mathcal{M}} \rightarrow \overline{\mathcal{N}}$ be a bijection. Let g be a Y -computable function such that for every basic ball B_q^i , $F(B_q^i) = \mathcal{O}(W_{g(i,q)}^X)$. Assume also that for every $x \in \overline{\mathcal{M}}$, $\inf \{ \text{diam}(F(B)) \mid x \in B \text{ and } B \text{ is a basic open ball} \} = 0$. Then F is $Y \oplus X''$ -effectively continuous.*

Proof. Given any basic open ball $\hat{B}$ of $\overline{\mathcal{N}}$ we use Y to enumerate the range of g , and for each element k in the range of g we use X'' to check if $\mathcal{O}(W_k^X) \subseteq^{\mathfrak{f}} \hat{B}$. This allows $Y \oplus X''$ to enumerate

the set $\{B_q^i \mid F(B_q^i) \subseteq^f \hat{B}\}$, uniformly in $\hat{B}$. By our assumption,

$$\inf \{\text{diam}(F(B)) \mid x \in B \text{ and } B \text{ is a basic open ball}\} = 0,$$

we conclude that $F^{-1}(\hat{B}) = \bigcup \{B_q^i \mid F(B_q^i) \subseteq^f \hat{B}\}$. $\square$

We shall also need the following definition in the proof of Theorem 4.12 and Theorem 4.14. In the proof of these two theorems we will need to define sets which are finite Boolean combinations of sets in $\mathcal{V}$. Since we do not require the effective clopen basis $\mathcal{V}$ to be closed under taking finite Boolean combinations, we will need to introduce a notation for such sets:

Definition 4.11. Let $\mathcal{V}$ be an effective clopen basis for an effectively zero-dimensional presentation $\mathcal{M}$ of the Baire space. A B -code for $\mathcal{V}$ is defined inductively by the following.

- A finite sequence of the form $\chi = \langle \cup, \langle e_0, k_0, j_0 \rangle, \dots, \langle e_N, k_N, j_N \rangle \rangle$ such that $\langle e_0, k_0 \rangle, \dots, \langle e_N, k_N \rangle \in \mathcal{V}$ and $j_0, \dots, j_N \in \{0, 1\}$ is a B -code. This code stands for the set $\mathcal{O}(W_{c_0}) \cup \dots \cup \mathcal{O}(W_{c_N})$ where $c_i = e_i$ if $j_i = 0$, and $c_i = k_i$ otherwise, for each $i \leq N$. Let $m(\chi)$ be a number m such that $\mathcal{O}(W_m) = \mathcal{O}(W_{c_0}) \cup \dots \cup \mathcal{O}(W_{c_N})$.
- A finite sequence of the form $\chi = \langle \cap, \langle e_0, k_0, j_0 \rangle, \dots, \langle e_N, k_N, j_N \rangle \rangle$ such that $\langle e_0, k_0 \rangle, \dots, \langle e_N, k_N \rangle \in \mathcal{V}$ and $j_0, \dots, j_N \in \{0, 1\}$ is a B -code. Similarly, let $m(\chi)$ be a number m such that $\mathcal{O}(W_m) = \mathcal{O}(W_{c_0}) \cap \dots \cap \mathcal{O}(W_{c_N})$ where $c_i = e_i$ if $j_i = 0$, and $c_i = k_i$ otherwise, for each $i \leq N$.
- If $\chi_0, \dots, \chi_N$ are B -codes, then so are the strings $\delta_0 = \langle \cup, \chi_0, \dots, \chi_N \rangle$ and $\delta_1 = \langle \cap, \chi_0, \dots, \chi_N \rangle$, and we let $m(\delta_0)$ and $m(\delta_1)$ be indices such that $\mathcal{O}(W_{m(\delta_0)}) = \mathcal{O}(W_{m(\chi_0)}) \cup \dots \cup \mathcal{O}(W_{m(\chi_N)})$ and $\mathcal{O}(W_{m(\delta_1)}) = \mathcal{O}(W_{m(\chi_0)}) \cap \dots \cap \mathcal{O}(W_{m(\chi_N)})$.

Obviously, the index $m(\chi)$ can always be found effectively from χ . Since each B -code records both open and co-open indices, the complementation operation can be performed effectively. That is, we can let $\langle \cup, \chi_0, \dots, \chi_N \rangle^c = \langle \cap, \chi_0^c, \dots, \chi_N^c \rangle$ and $\langle \cap, \chi_0, \dots, \chi_N \rangle^c = \langle \cup, \chi_0^c, \dots, \chi_N^c \rangle$, and $\langle e, k, j \rangle^c = \langle e, k, 1-j \rangle$. By convention, we take $m(\langle \cup, \emptyset \rangle)$ to be some open index for $\overline{\mathcal{M}}$, and $m(\langle \cap, \emptyset \rangle) = m(\langle \cup, \emptyset \rangle^c)$ to be some open index for $\emptyset$. For any B -code χ , we have that $\mathcal{O}(W_{m(\chi)}), \mathcal{O}(W_{m(\chi^c)})$ form a partition of $\overline{\mathcal{M}}$. These definitions can all be relativized to an oracle X .

We are now ready to prove:

Theorem 4.12. *Let X be any oracle and $\mathcal{M}$ and $\mathcal{N}$ be X -computable presentations of the Baire space that are X -effectively zero-dimensional. Then there is an X'' -computable homeomorphism between $\overline{\mathcal{M}}$ and $\overline{\mathcal{N}}$.*

Proof. It is enough to fix $\mathcal{M}$ and take $\mathcal{N}$ to be the standard copy $\omega^{<\omega}$. Fix the X -c.e. clopen basis $\mathcal{V}^X$ for $\overline{\mathcal{M}}$. We will analyze the complexity of the proof of the Alexandrov-Urysohn Theorem (see [Kec95, Theorem 7.7]). We define the Lusin scheme $\langle \chi_\sigma \rangle_{\sigma \in \omega^{<\omega}}$ on $\overline{\mathcal{M}}$ as follows. Each χ_σ will be a B -code for $\mathcal{V}^X$ (see Definition 4.11, above).

Let $\chi_\emptyset = \langle \cup, \emptyset \rangle$, a B -code for $\overline{\mathcal{M}}$. Now assume that χ_σ is given such that $\mathcal{O}(W_{m(\chi_\sigma)}^X) \neq \emptyset$. Since $\mathcal{O}(W_{m(\chi_\sigma)}^X)$ is non-empty and open, it is not compact as all compact subsets of (any copy of) the Baire space have empty interior. Since it is also closed, it cannot be totally bounded and so there is some $\varepsilon < 2^{-|\sigma|}$ such that no finite cover of $\mathcal{O}(W_{m(\chi_\sigma)}^X)$ by open sets of diameter less than ε exist.

Now search for some dyadic rational $\varepsilon < 2^{-|\sigma|}$ satisfying the property:

“For every finite collection of B -codes for $\mathcal{V}^X$, $\chi_0, \dots, \chi_N$, where each $\mathcal{O}(W_{m(\chi_i)}^X)$ has diameter less than ε , we have that $\mathcal{O}(W_{m(\chi_\sigma)}^X) \cap \mathcal{O}(W_{m(\chi_0)}^X) \cap \dots \cap \mathcal{O}(W_{m(\chi_N)}^X) \neq \emptyset$ ”.

As the property above is $\Pi_2^0(X)$ for each value of ε , this search can be carried out by X'' .

Now having obtained ε we fix an enumeration $\langle d_i, l_i \rangle_{i \in \omega}$ of all elements of $\mathcal{V}^X$ such that $\text{diam}(\mathcal{O}(W_{d_i}^X)) < \frac{\varepsilon}{3}$ for all i . Assume inductively that $\chi_{\sigma*i}$ have been defined for all $i < N$. We let $\chi_{\sigma*N}$ be determined in the following way. For the first j such that $\mathcal{O}(W_{d_j}^X) \cap \mathcal{O}(W_{m(\chi_\sigma)}^X) \cap \bigcap_{i < N} \mathcal{O}(W_{m(\chi_{\sigma*i}^c)}^X) \neq \emptyset$,

we take $\chi_{\sigma*N} = \langle \cap, \chi_\sigma, \chi_{\sigma*0}^c, \dots, \chi_{\sigma*(N-1)}^c, \langle d_j, l_j, 0 \rangle \rangle$.

The definition of $\langle \chi_\sigma \rangle_{\sigma \in \omega^{<\omega}}$ is X'' -computable. We can verify by induction on $|\sigma|$ that for each σ and N ,

- (i) $\mathcal{O}(W_{m(\chi_{\sigma*N})}^X) \neq \emptyset$ and has diameter less than $\varepsilon < 2^{-|\sigma|}$ (note that ε depends on σ).
- (ii) $\mathcal{O}(W_{m(\chi_{\sigma*0})}^X), \mathcal{O}(W_{m(\chi_{\sigma*1})}^X), \dots$ are all defined and form a partition of $\mathcal{O}(W_{m(\chi_\sigma)}^X)$.

To see (i), first note that by the inductive hypothesis, $\mathcal{O}(W_{m(\chi_\sigma)}^X) \neq \emptyset$, and therefore ε can be found at this step. We may assume that each open ball enumerated into $\mathcal{O}(W_{m(\chi_{\sigma*N})}^X)$ is formally contained in some ball of $\mathcal{O}(W_{d_j}^X)$. By the definition of the formal diameter, we see that $\text{diam}(\mathcal{O}(W_{m(\chi_{\sigma*N})}^X)) \leq 3 \cdot \text{diam}(\mathcal{O}(W_{d_j}^X)) < \varepsilon$. We now explain why (ii) holds. Suppose that $\chi_{\sigma*i}$ are defined for all $i < N$. By (i), the formal diameter of $\mathcal{O}(W_{m(\chi_{\sigma*i})}^X) < \varepsilon$ for all $i < N$. Therefore, by the property of ε , $\mathcal{O}(W_{m(\chi_\sigma)}^X) \cap \bigcap_{i < N} \mathcal{O}(W_{m(\chi_{\sigma*i}^c)}^X) \neq \emptyset$. Since $\mathcal{V}^X$ encodes a basis, we must be able to find a candidate for $\chi_{\sigma*N}$. Finally, the fact that the sequence of open sets coded by $\chi_{\sigma*0}, \chi_{\sigma*1}, \dots$ form a partition follows from the construction.

Now let $F : \omega^\omega \rightarrow \overline{\mathcal{M}}$ by taking $F(x) = \lim_{n \in \omega} \beta_n$ where β_n is the first special point enumerated in $\mathcal{O}(W_{m(\chi_{x|n})}^X)$ for every n . Then $F(x)$ is well-defined for every x by property (i), and is bijective because of (ii) and the fact that $\mathcal{O}(W_{m(\chi_\emptyset)}^X) = \overline{\mathcal{M}}$. F is X'' -effectively open because $F([\sigma]) = \mathcal{O}(W_{m(\chi_\sigma)}^X)$ for every $\sigma \in \omega^{<\omega}$.

Now since $\sigma \mapsto m(\chi_\sigma)$ is X'' -computable, we apply Lemma 4.10 to conclude that F is X'' -effectively continuous. $\square$

Corollary 4.13. *Any two effectively zero-dimensional presentations of the Baire space are $\mathbf{0}''$ -homeomorphic.*

We now turn to the problem of computing a lowerbound on the categoricity of the Baire space. Again, this seems to depend critically on the ability to locate clopen sets in a given presentation, so we limit our question to consider only effectively zero-dimensional copies. By Corollary 4.13, $\mathbf{0}''$ is an upperbound for effectively zero-dimensional copies, while by Theorem 4.8, $\mathbf{0}'$ is not an upperbound. This might lead us to believe that the degree of categoricity of the Baire space, when restricted to effectively zero-dimensional copies, is $\mathbf{0}''$. To prove this conjecture, we might want to *encode* $\emptyset''$ into a pair of effectively zero-dimensional presentations $\overline{\mathcal{M}}$ and $\overline{\mathcal{N}}$ of the Baire space, so that any homeomorphism between $\overline{\mathcal{M}}$ and $\overline{\mathcal{N}}$ can compute $\emptyset''$. Usually (especially in computable structure theory) one can achieve this by taking a diagonalization strategy, such as the proof of Theorem 4.8, and turn that proof into one that encodes $\emptyset''$, or at least $\emptyset'$, into homeomorphisms between $\overline{\mathcal{M}}$ and ω^ω . However, we will show that the intuition from computable structure theory fails here. Not only are we unable to encode $\emptyset''$ or $\emptyset'$ into a pair of effectively zero-dimensional presentations of the Baire space, we cannot even encode any non-computable information (Theorem 4.14). As a consequence, we conclude that the degree of categoricity for zero-dimensional copies is not well-defined (Corollary 4.18). However, it is still possible that the degree of categoricity for the Baire space exists, but we

conjecture that by using the methods in this paper, one can extend the results to all computable copies of the Baire space. We leave this (Question 4.19) open.

Theorem 4.14. *For each non-computable set X there is some $A \not\leq_T X$ such that any two effectively zero-dimensional presentations of the Baire space are A -homeomorphic.*

Proof. We fix a non-computable set X and define a sequence $\beta_0 \subset \beta_1 \subset \dots$ of initial segments of A . At the end we take $A = \bigcup_s \beta_s$. As usual we let $\{\beta_s\}$ be finite binary sequences and write $D(\beta) = \{n \in \omega \mid \beta(n) = 1\}$. Let $\langle \mathcal{M}_c, \mathcal{V}_c \rangle_{c \in \omega}$ be a computable listing of all pairs $\langle \mathcal{M}, \mathcal{V} \rangle$ such that $\mathcal{M} = (\{\alpha_n\}_{n \in \omega}, d)$ is a partial computable non-empty Polish structure and $\mathcal{V}$ is a c.e. set. To prove the theorem it suffices to show that for each c such that $\mathcal{M}_c$ is an effectively zero-dimensional presentation of the Baire space with an effective clopen basis encoded by $\mathcal{V}_c$ (we shorten this by simply saying $\langle \mathcal{M}_c, \mathcal{V}_c \rangle$ is *effectively zero-dimensional*), there is an A -computable homeomorphism between ω^ω and $\overline{\mathcal{M}_c}$. Obviously, not every pair in the list $\langle \mathcal{M}_c, \mathcal{V}_c \rangle_{c \in \omega}$ is an effectively zero-dimensional pair, and we shall not be interested in every pair in this list. However, during the construction there will be certain actions which will rely on a “computable search” (see the definition of being E_s -correct below), and for the sake of these actions we are forced to consider a computable list of all potential pairs.

4.1. Notation used. We will identify each element of A with the tuple $\langle c, n, \sigma, \chi \rangle$ where $c, n \in \omega$ and $\sigma, \chi \in \omega^{<\omega}$. The intended meaning of $\langle c, n, \sigma, \chi \rangle \in A$ is that the oracle A encodes that $F_c([\sigma]) = \mathcal{O}(W_{m(\chi)})$, where χ is a B -code for $\mathcal{V}_c$. (We refer the reader to Definition 4.11 and the proof of Theorem 4.12; we will utilize ideas from there). Here F_c will be some homeomorphism from ω^ω to $\overline{\mathcal{M}_c}$. The number n is to allow the oracle A the possibility of never deciding on $F_c([\sigma])$, due to the possibility of c being an incorrect index. The actual definition of F_c will be given later in the proof. The oracle A will map each basic open cylinder $[\sigma]$ of ω^ω to some effectively clopen set in $\overline{\mathcal{M}_c}$.

We shall also require the function E , where at each step s of the construction we will only have decided a finite subset of its domain. The intention is that $E(c, \sigma)$ represents the value of ε in the proof of Theorem 4.12, specifically when considering the pair $\langle \mathcal{M}_c, \mathcal{V}_c \rangle$ and at the step where $\chi_{\sigma * N}$ is being defined. Given a finite function E_s (the partial approximation to E at stage s) and a finite set D , we say that D is E_s -correct if the following conditions are met:

- (I) For every $\langle c, n, \sigma, \chi \rangle \in D$, we have that χ is a B -code for $\mathcal{V}_c$ and $\mathcal{O}(W_{m(\chi)}) \neq \emptyset$.
- (II) For each c and σ , there is at most one n and one χ such that $\langle c, n, \sigma, \chi \rangle \in D$.
- (III) If $\langle c, n_{\sigma * N}, \sigma * N, \chi_{\sigma * N} \rangle \in D$, then $\langle c, n_\sigma, \sigma, \chi_\sigma \rangle \in D$ and $\langle c, n_{\sigma * i}, \sigma * i, \chi_{\sigma * i} \rangle \in D$ for some collection of such tuples, for all $i < N$. Furthermore we have:
 - (i) $\chi_{\sigma * N} = \left\langle \bigcap, \chi_\sigma, \chi_{\sigma * 0}^c, \dots, \chi_{\sigma * (N-1)}^c, \langle d, l, 0 \rangle \right\rangle$ for some d, l .
 - (ii) $\mathcal{O}(W_{m(\chi_\sigma)}) \cap \mathcal{O}(W_{m(\chi_{\sigma * 0}^c)}) \cap \dots \cap \mathcal{O}(W_{m(\chi_{\sigma * N}^c)}) \neq \emptyset$.
 - (iii) If $E_s(c, \sigma) \downarrow > \frac{1}{N}$, we have that $\text{diam}(\mathcal{O}(W_d)) < \frac{1}{3}E(c, \sigma)$.
 - (iv) $\text{diam}(\mathcal{O}(W_d)) < \frac{1}{3}2^{-|\sigma|}$.

Notice that the listing $\langle \mathcal{M}_c, \mathcal{V}_c \rangle_{c \in \omega}$ is computable, and the formal diameter of clopen sets in $\mathcal{V}_c$ are given uniformly partial computably. If we are also given the canonical indices of D and of $\text{dom}(E_s)$, then checking if D is E_s -correct is Σ_1^0 . Now given a finite set D , c, σ and χ , we say that $D \llbracket c, \sigma \rrbracket = \chi$ if $\langle c, n, \sigma, \chi' \rangle \in D$ for some such tuple, and $\chi' = \chi$ for the least such n . Finally, notice that the function m and the open set $\mathcal{O}(W_{m(\chi)})$ depend on the structure $\mathcal{M}_c$, and so it is technically more correct to write m_c and $\mathcal{O}_c$ instead of simply m and $\mathcal{O}$; however we suppress the index c from our notation.

4.2. Construction. We construct the sequence $\beta_0 \subset \beta_1 \subset \dots$ of finite initial segments of A and $E_0 \subseteq E_1 \subseteq \dots$, a sequence of finite functions. At $s = 0$ we set $\beta_0 = E_0 = \emptyset$. Now assume that we are given the finite string β_s and the finite function E_s . Let $D = D(\beta_s)$. There are two cases.

First suppose that $s = 2t$. We wish to satisfy the requirement $X \neq \Phi_t^A$. We consider the following property:

- (†) There is some extension $\beta \supset \beta_s$ and some n such that $D(\beta)$ is E_s -correct, and $\Phi_t^\beta(n) \downarrow \neq X(n)$.

If property (†) holds then we take $\beta_{s+1} = \beta$ where β is the first string found satisfying (†), and $E_{s+1} = E_s$ (we will henceforth assume that if a parameter is not updated at a step then it retains its previous value). If property (†) does not hold then we take $\beta_{s+1} = \beta * 0$.

Now suppose that $s = 2\langle m, c, \sigma \rangle + 1$. At this stage we act for the pair $\langle c, \sigma \rangle$. There are several possibilities. First of all, if $\sigma = \emptyset$ and $m = 0$ then we set $\beta_{s+1} \supset \beta_s$ to be the least string such that $D(\beta_{s+1}) = D \cup \{\langle c, n, \emptyset, \langle \cup, \emptyset \rangle \rangle\}$ for some n . Otherwise, if $D \llbracket c, \sigma \rrbracket \uparrow$ we do nothing at this step. Thus we can assume that $D \llbracket c, \sigma \rrbracket \downarrow$, and let N be the least number such that $D \llbracket c, \sigma * N \rrbracket \uparrow$. There are two further subcases:

- (i) $E_s(c, \sigma) \downarrow$: Let $\langle d, l \rangle \in \mathcal{V}_c$ be the first pair found such that $\text{diam}(\mathcal{O}(W_d)) < \frac{1}{3}E_s(c, \sigma)$, and where

$$\mathcal{O}(W_d) \cap \mathcal{O}(W_{m(D \llbracket c, \sigma \rrbracket)}) \cap \bigcap_{i < N} \mathcal{O}(W_{m(D \llbracket c, \sigma * i \rrbracket^c)}) \neq \emptyset.$$

We take $\chi = \langle \cap, D \llbracket c, \sigma \rrbracket, D \llbracket c, \sigma * 0 \rrbracket^c, \dots, D \llbracket c, \sigma * (N-1) \rrbracket^c, \langle d, l, 0 \rangle \rangle$, and take $\beta_{s+1} \supset \beta_s$ to be the least string such that $D(\beta_{s+1}) = D \cup \{\langle c, n, \sigma * N, \chi \rangle\}$ for some n . If $\langle d, l \rangle$ does not exist, do nothing.

- (ii) $E_s(c, \sigma) \uparrow$: Let $\varepsilon < 2^{-|\sigma|}$ be some dyadic rational number satisfying the property:

“For every finite collection of B -codes for $\mathcal{V}_c$, $\chi_0, \dots, \chi_{N'}$, where each $\mathcal{O}(W_{m(\chi_i)})$ has diameter less than ε , we have that $\mathcal{O}(W_{m(D \llbracket c, \sigma \rrbracket)}) \cap \bigcap_{i < N} \mathcal{O}(W_{m(D \llbracket c, \sigma * i \rrbracket^c)}) \cap \mathcal{O}(W_{m(\chi_0^c)}) \cap \dots \cap \mathcal{O}(W_{m(\chi_{N'}^c)}) \neq \emptyset$ ”.

If ε exists, we take $E_{s+1}(c, \sigma) \downarrow = \varepsilon$. Follow the instructions above in (i) to determine the values of $\chi_{\sigma * N}, \dots, \chi_{\sigma * M}$, corresponding respectively to $\sigma * N, \dots, \sigma * M$, where M is the least such that $E_s(c, \sigma) \downarrow > \frac{1}{M}$. Take β_{s+1} to be the least string such that $D(\beta_{s+1}) = D \cup \{\langle c, n_{\sigma * N}, \sigma * N, \chi_{\sigma * N} \rangle, \dots, \langle c, n_{\sigma * M}, \sigma * M, \chi_{\sigma * M} \rangle\}$ for some such tuples.

At the end of the construction we take $A = \cup_s \beta_s$.

4.3. Verification. We now verify that the construction works.

Lemma 4.15. *At every stage s of the construction, $D(\beta_s)$ is E_s -correct.*

Proof. At stage $s = 0$, $D(\beta_s) = E_s = \emptyset$ and (I), (II) and (III) hold vacuously. Now assume that $D = D(\beta_s)$ is E_s -correct. If s is even then as $E_{s+1} = E_s$, and $\beta_{s+1} = \beta_s * 0$ or some β satisfying (†), obviously the property holds for $s + 1$. So we assume that $s = 2\langle m, c, \sigma \rangle + 1$. We check each case considered in the construction. If we did nothing at this step s then the statement for $s + 1$ obviously holds, so we assume some action was taken.

If $\sigma = \emptyset$ and $m = 0$ then $D(\beta_{s+1}) = D \cup \{\langle c, n, \emptyset, \langle \cup, \emptyset \rangle \rangle\}$ for some n and $E_{s+1} = E_s$. Then (I) holds as $\langle \cup, \emptyset \rangle$ is a B -code for $\mathcal{V}_c$ and $\overline{\mathcal{M}}_c \neq \emptyset$ by assumption. (II) holds since $m = 0$ and so s is the first stage we have ever acted for the pair $\langle c, \sigma \rangle$. As $\sigma = \emptyset$, (III) holds vacuously.

Assume that $D \llbracket c, \sigma \rrbracket \downarrow$ and N is the least number such that $D \llbracket c, \sigma * N \rrbracket \uparrow$, and $E_s(c, \sigma) \uparrow$. Then $\text{dom}(E_{s+1}) = \text{dom}(E_s) \cup \{(c, \sigma)\}$. (Note that we are not claiming that the value of ε can always be found at every such stage; indeed, if $\mathcal{M}_c$ is not a copy of the Baire space then we may do nothing

at this stage; we merely assume that s is an odd stage where the construction did manage to find $\varepsilon = E_{s+1}(c, \sigma)$.

To $D = D(\beta_s)$ we add the tuples $\langle c, n_{\sigma*N}, \sigma * N, \chi_{\sigma*N} \rangle, \dots, \langle c, n_{\sigma*M}, \sigma * M, \chi_{\sigma*M} \rangle$ where M is the least such that $E_s(c, \sigma) > \frac{1}{M}$. Each of these tuples is decided according to the instructions in subcase (i) of the construction, and it is easy to verify that (I) holds for each of these tuple $\langle c, n_{\sigma*i}, \sigma * i, \chi_{\sigma*i} \rangle$, where $N \leq i \leq M$. To check that (II) holds for each new tuple $\langle c, n_{\sigma*i}, \sigma * i, \chi_{\sigma*i} \rangle$, note that as $i \geq N$, by the inductive hypothesis, $D\llbracket c, \sigma * i \rrbracket \uparrow$ at the beginning of stage s . Property (III)(i) is ensured by the construction. To see property (III)(ii), suppose that $\mathcal{O}(W_{m(D\llbracket c, \sigma \rrbracket)}) \cap \bigcap_{i < N} \mathcal{O}(W_{m(D\llbracket c, \sigma*i \rrbracket^c)}) \cap \bigcap_{N \leq i \leq M} \mathcal{O}(W_{m(\chi_{\sigma*i}^c)}) = \emptyset$. Then this would contradict the property of ε as each $\mathcal{O}(W_{m(\chi_{\sigma*i})})$ has formal diameter no more than $3 \cdot \text{diam}(\mathcal{O}(W_d)) < \varepsilon$. (See the proof of Theorem 4.12 for a similar calculation). Property (III)(iii) holds since only M applies. (III)(iv) holds as $\varepsilon < 2^{-|\sigma|}$.

Now assume that $D\llbracket c, \sigma \rrbracket \downarrow$ and N is the least number such that $D\llbracket c, \sigma * N \rrbracket \uparrow$, and $E_s(c, \sigma) \downarrow$. Then $E_{s+1} = E_s$. To $D = D(\beta_s)$ we only add the tuple $\langle c, n, \sigma * N, \chi \rangle$ with χ and n determined by the construction. Again, (I) and (II) obviously hold for the tuple $\langle c, n, \sigma * N, \chi \rangle$. (III)(i), (iii) and (iv) are ensured by the construction and the fact that $E(c, \sigma) < 2^{-|\sigma|}$. Now we check Property (III)(ii). Let $t < s$ be the stage where $E(c, \sigma) = \varepsilon$ is defined and let N_t be the value of N at stage t . At stage t we ensured that $\mathcal{O}(W_{m(D\llbracket c, \sigma*i \rrbracket)})$ is defined and has formal diameter less than ε for each value of $N_t \leq i \leq \lceil \frac{1}{\varepsilon} \rceil$. Now as D is E_s -correct, we also have the same for all i satisfying $\lceil \frac{1}{\varepsilon} \rceil < i < N$, and for $i = N$ the diameter is also bounded by the construction. Therefore, by the property of ε we have $\mathcal{O}(W_{m(D\llbracket c, \sigma \rrbracket)}) \cap \bigcap_{i < N_t} \mathcal{O}(W_{m(D\llbracket c, \sigma*i \rrbracket^c)}) \cap \bigcap_{N_t \leq i \leq N} \mathcal{O}(W_{m(D\llbracket c, \sigma*i \rrbracket^c)}) \neq \emptyset$, giving us (III)(ii). $\square$

We let $A = \cup_s \beta_s$. First we prove:

Lemma 4.16. $A \not\leq_T X$.

Proof. Suppose there is some t such that $\Phi_t^A = X$. Since $\beta_{2t+1} \subset A$, at stage $s = 2t$, property $(\dagger)$ does not hold. Now we argue that X is computable. For each n search for the first $\gamma_n \supset \beta_s$ such that $D(\gamma_n)$ is E_s -correct and such that $\Phi_t^{\gamma_n}(n) \downarrow$. Since Φ_t^A is total there is some $s' > s$ such that $\Phi_t^{\beta_{s'}}(n) \downarrow$ and where $D(\beta_{s'})$ is $E_{s'}$ -correct (by Lemma 4.15), and is therefore also E_s -correct. Hence the search for γ_n must terminate, and since checking for correctness is Σ_1^0 , the value of $\Phi_t^{\gamma_n}(n)$ can be found effectively in n . Since property $(\dagger)$ does not hold, we conclude that $\Phi_t^{\gamma_n}(n) = X(n)$, contradicting the fact that X is not computable. $\square$

Lemma 4.17. Any two effectively zero-dimensional presentations of the Baire space are A -homeomorphic.

Proof. Let $\langle \mathcal{M}_c, \mathcal{V}_c \rangle$ be effectively zero-dimensional. First of all note that by the correctness of $D(\beta_s)$, for each σ there is at most one tuple $\langle c, n, \sigma, \chi \rangle \in D(\beta_s)$ for each s . We let χ_σ be the unique value of χ if some such tuple $\langle c, n, \sigma, \chi \rangle$ appears in $D(\beta_s)$ for some s .

We follow the proof of Theorem 4.12. We check that given any σ and N , the following hold:

- (i) $E(c, \sigma)$, χ_σ and $\chi_{\sigma*N}$ are eventually defined, $\mathcal{O}(W_{m(\chi_{\sigma*N})}) \neq \emptyset$ and has diameter less than $2^{-|\sigma|}$.
- (ii) $\mathcal{O}(W_{m(\chi_{\sigma*0})}), \mathcal{O}(W_{m(\chi_{\sigma*1})}), \dots$ form a partition of $\mathcal{O}(W_{m(\chi_\sigma)})$.

We assume that the claim above fails for some minimal string σ and some least N for σ . First of all observe that $\chi_\sigma \downarrow$, since if $|\sigma| > 0$ we can apply the minimality of σ , and if $\sigma = \emptyset$ the construction must define $\chi_\emptyset = \langle \cup, \emptyset \rangle$ at stage $s = 2 \langle 0, c, \emptyset \rangle + 1$. By Lemma 4.15, $D(\beta_s)$ is correct (in particular, (III)(ii) holds) for all sufficiently large s , and by the minimality of N , we see that

$\mathcal{O}(W_{m(\chi_\sigma)}) \cap \bigcap_{i < N} \mathcal{O}(W_{m(\chi_{\sigma * i}^c)}) \neq \emptyset$. Since $\langle \mathcal{M}_c, \mathcal{V}_c \rangle$ is effectively zero-dimensional and the aforementioned set is non-empty and clopen (by item (I) of correctness) this means that $\mathcal{O}(W_{m(\chi_\sigma)}) \cap \bigcap_{i < N} \mathcal{O}(W_{m(\chi_{\sigma * i}^c)})$ is not totally bounded. Hence, $E(c, \sigma)$ cannot stay undefined forever.

This means that for a large enough odd stage s we will act under case (i) of the construction to define $D(\beta_{s+1}) \llbracket c, \sigma * N \rrbracket$. As $\mathcal{O}(W_{m(\chi_\sigma)}) \cap \bigcap_{i < N} \mathcal{O}(W_{m(\chi_{\sigma * i}^c)}) \neq \emptyset$ and $\mathcal{V}_c$ is a basis for $\overline{\mathcal{M}}_c$, we must succeed in finding $\langle d, l \rangle \in \mathcal{V}_c$ at stage s . Hence $\chi_{\sigma * N}$ will be defined after all. By correctness, $\mathcal{O}(W_{m(\chi_{\sigma * N})}) \neq \emptyset$ and has diameter less than $2^{-|\sigma|}$ will hold as long as $\chi_{\sigma * N} \downarrow$.

Therefore (ii) must fail for σ . By item (III)(i) of correctness, the sequence $\mathcal{O}(W_{m(\chi_{\sigma * 0})}), \mathcal{O}(W_{m(\chi_{\sigma * 1})}), \dots$ is mutually disjoint. We also have $\mathcal{O}(W_{m(\chi_\sigma)}) \subseteq \bigcup_i \mathcal{O}(W_{m(\chi_{\sigma * i})})$ because each pair $\langle d, l \rangle \in \mathcal{V}_c$ of formal diameter less than $\frac{1}{3}E(c, \sigma)$ is given infinitely many chances to be included (specifically, at all stages of the form $2 \langle m, c, \sigma \rangle + 1$). This gives the desired contradiction.

Note that $\chi_\emptyset = \langle \cup, \emptyset \rangle$, a B -code for $\overline{\mathcal{M}}_c$. Now we argue exactly as in Theorem 4.12. Let $F_c : \omega^\omega \rightarrow \overline{\mathcal{M}}$ by taking $F_c(x) = \lim_{n \in \omega} \beta_n$ where β_n is the first special point enumerated in $\mathcal{O}(W_{m(\chi_{x \upharpoonright n})})$ for every n . Then $F_c(x)$ is well-defined for every x by property (i), and is bijective because of (ii) and the fact that $\mathcal{O}(W_{m(\chi_\emptyset)}) = \overline{\mathcal{M}}$. F_c is A -effectively open because $F_c([\sigma]) = \mathcal{O}(W_{m(\chi_\sigma)})$ for every $\sigma \in \omega^{<\omega}$.

Just like in the proof of Theorem 4.12, we wish to now apply Lemma 4.10 to conclude that F_c is continuous. Unfortunately, Lemma 4.10 only allows us to conclude that F_c is $A \oplus \emptyset''$ -effectively continuous. We shall have to modify the proof of Lemma 4.10 to work for effectively zero-dimensional presentations.

The map $\sigma \mapsto \langle m(\chi_\sigma), \mu(\chi_\sigma) \rangle$ is A -computable and total, where $\mu(\chi_\sigma)$ is a computable approximation to $3 \cdot \text{diam}(\mathcal{O}(W_d))$ and where χ_σ is of the form $\langle \cap, \dots, \langle d, l, 0 \rangle \rangle$ (by correctness). Fix an open ball $\hat{B}$ of $\overline{\mathcal{M}}_c$. Consider the set D of all open cylinders $[\sigma]$ such that if α_i is the first special point to be enumerated in $\mathcal{O}(W_{m(\chi_\sigma)})$ then $B_{2\mu(\chi_\sigma)}^i \subseteq^f \hat{B}$. Clearly D is A -c.e., uniformly in $\hat{B}$. Now we claim that $\bigcup D = F_c^{-1}(\hat{B})$. First, note that as $\text{diam}(\mathcal{O}(W_{m(\chi_\sigma)})) \leq 3 \cdot \text{diam}(\mathcal{O}(W_d)) = \mu(\chi_\sigma)$, we see that if $[\sigma] \in D$ then $\mathcal{O}(W_{m(\chi_\sigma)}) \subseteq B_{2\mu(\chi_\sigma)}^i \subseteq \hat{B}$. Hence $\bigcup D \subseteq F_c^{-1}(\hat{B})$. The reverse containment follows from the fact that $\mathcal{O}(W_{m(\chi_\sigma)})$ has formal diameter less than $2^{-|\sigma|+1}$. Hence F_c is A -effectively continuous. $\square$

This ends the proof of Theorem 4.14. $\square$

Theorem 4.14 tells us that for each non-computable X , there is no pair of effectively zero-dimensional presentations $\overline{\mathcal{M}}$ and $\overline{\mathcal{N}}$ of the Baire space such that every homomorphism between $\overline{\mathcal{M}}$ and $\overline{\mathcal{N}}$ computes X . Hence we cannot encode any non-computable information into zero-dimensional presentations in this sense.

Corollary 4.18. *When restricted to effectively zero-dimensional presentations, the degree of categoricity of the Baire space does not exist.*

Proof. If the degree of categoricity for effectively zero-dimensional presentations exist, say $\mathbf{x}$, then by Theorem 4.8, $\mathbf{x} \neq \mathbf{0}$. By Theorem 4.14, $\mathbf{x}$ cannot be the least degree. $\square$

While we have restricted our study to effectively zero-dimensional presentations of the Baire space, it is still possible that non-computable information can be encoded into a pair of computable presentations of the Baire space. Thus we leave open the question:

Question 4.19. *Does the Baire space have a degree of categoricity?*

5. COMPUTABLY COMPACT SPACES

As we already noted in the introduction, (3) of Theorem 4.8, follows from Lemma 2.5 and the following:

Theorem 5.1. *If $\mathcal{X}$ is compact computable Polish, then there exists $\mathcal{Y} \cong \mathcal{X}$ computable Polish such that $\mathcal{Y}$ is not computably compact.*

Let $\mathcal{X} = \overline{(\{x_i\}_{i \in \omega}, d)}$, and consider the embedding which maps each point x to the vector $(v_i)_{i \in \omega} \in \mathcal{H}$ such that $v_{2i+1} = 0$ and $v_{2i} = d(x, x_i)$ for each $i \in \omega$. Since $\mathcal{X}$ is compact, this is clearly a homeomorphism and we may thus assume that $\mathcal{X} = \overline{(\{x_i\}_{i \in \omega}, \|\cdot\|_{\mathcal{H}})} \subseteq \mathcal{H}$.

The goal is to construct a space $\mathcal{Y} = \overline{(\{y_i\}_{i \in \omega}, \|\cdot\|_{\mathcal{H}})} \subseteq \mathcal{H}$ satisfying the following requirements:

R_e : There exists n such that $\Phi_e(n)$ is not (the code of) a finite 2^{-n} cover of $\mathcal{Y}$.

The intuitive idea to satisfy each requirement is simple. We wait for $\Phi_e(n) \downarrow$ for some e, n , and it must produce (the code of) some finite collection of special points of $\mathcal{Y}$, say $y_{i_0}, y_{i_1}, \dots, y_{i_m}$. Once such a collection is found, we have to ensure that $\bigcup_{j \leq m} B(y_{i_j}, 2^{-n})$ is not a cover of $\mathcal{Y}$; fix the positions of y_{i_j} for each $j \leq m$, and define the special point with a fresh index to be of distance at least 2^{-n} away from each of y_{i_j} . The main tension is obviously to perform the strategy as described for each requirement while keeping $\mathcal{Y}$ compact.

5.1. Construction. During the construction, we define functions f_s for each $s \in \omega$. Initially set $f_0 : \mathcal{X} \rightarrow \mathcal{H}$ to be the identity function. At each stage s , let $e < s$ be the least such that $\Phi_e(2e+1)[s] \downarrow$ and R_e has not *acted* (to be defined shortly). If no such e exists, define $f_{s+1} = f_s$, $y_s = f_{s+1}(x_s)$, then proceed to stage $s+1$.

Otherwise, let $y_s = f_{s+1}(x_s)$ where $f_{s+1} : \mathcal{X} \rightarrow \mathcal{H}$ is defined recursively as follows. f_0 is the identity function, and $f_{s+1}(x)$ adds $\max\{0, 1 - \varepsilon^{-1} \|f_s(x) - f_s(x_s)\|_{\mathcal{H}}\}$ to the $2e+1$ -th coordinate of $f_s(x) \in \mathcal{H}$, and ε is chosen such that $\varepsilon < \|f_s(x_n) - f_s(x_s)\|_{\mathcal{H}}$ for each $n \leq s$. Note that as long as $\|f_s(x_n) - f_s(x_s)\|_{\mathcal{H}} > 0$ for any $n < s$ (which we shall show in Lemma 5.2), then such an $\varepsilon \in \mathbb{Q}^+$ can be found effectively. Once ε has been found and f_{s+1} has been defined, we say that R_e has acted and proceed to stage $s+1$.

5.2. Verification. We now verify that the construction works; $\overline{(\{x_i\}_{i \in \omega}, \|\cdot\|_{\mathcal{H}})} \cong \overline{(\{y_i\}_{i \in \omega}, \|\cdot\|_{\mathcal{H}})}$, and that each R_e is satisfied. Without loss of generality, we may assume that $\|x_i - x_j\|_{\mathcal{H}} = 0$ iff $i = j$ since the “=” relation is co-c.e. for computable reals.

Lemma 5.2. *For each $s \in \omega$, $\|f_s(x) - f_s(y)\|_{\mathcal{H}} \geq \|x - y\|_{\mathcal{H}}$.*

Proof. Since f_0 is the identity function, we trivially have that $\|f_0(x) - f_0(y)\|_{\mathcal{H}} \geq \|x - y\|_{\mathcal{H}}$. Suppose inductively that $\|f_s(x) - f_s(y)\|_{\mathcal{H}} \geq \|x - y\|_{\mathcal{H}}$ for some $s \in \omega$. Clearly, if during stage s , no requirement acts, then $f_{s+1} = f_s$ and thus $\|f_{s+1}(x) - f_{s+1}(y)\|_{\mathcal{H}} = \|f_s(x) - f_s(y)\|_{\mathcal{H}} \geq \|x - y\|_{\mathcal{H}}$.

Suppose that R_e acts during stage s . Since $\|f_s(x) - f_s(y)\|_{\mathcal{H}} \geq \|x - y\|_{\mathcal{H}}$, it follows that $\varepsilon \in \mathbb{Q}^+$ such that $\varepsilon < \min_{n < s} \|f_s(x_n) - f_s(x_s)\|_{\mathcal{H}}$ exists and therefore may be found effectively as claimed (in the construction). From the construction, we also have that $f_{s+1}(x)$ is the vector

$$f_{s+1}(x)(i) = \begin{cases} f_s(x)(i) + \max\{0, 1 - \varepsilon^{-1} \|f_s(x) - f_s(x_s)\|_{\mathcal{H}}\} & \text{if } i = 2e + 1 \\ f_s(x)(i) & \text{otherwise.} \end{cases}$$

Recall that for each $x \in \mathcal{X} \subseteq \mathcal{H}$, $x(2i+1) = 0$. Furthermore, as each requirement R_e acts at most once, and possibly affects only the $2e+1$ -th coordinate of each vector in $\mathcal{X}$ when it acts,

we have that $|f_{s+1}(x)(2e+1) - f_{s+1}(y)(2e+1)| \geq |f_s(x)(2e+1) - f_s(y)(2e+1)| = 0$. Thus, $\|f_{s+1}(x) - f_{s+1}(y)\|_{\mathcal{H}} \geq \|f_s(x) - f_s(y)\|_{\mathcal{H}}$. $\square$

For each $e \in \omega$, let s_e be the stage (if it exists) during which R_e acts. From the definition of f_s , we have the following fact.

Fact 5.3. *If s_e exists, then for any $x \in \mathcal{X}$, $\|f_{s_e+1}(x) - f_{s_e}(x)\|_{\mathcal{H}} < 2^{-e}$.*

Thus, f_s converges uniformly to $f = \lim_s f_s$. Additionally, by applying Lemma 5.2, we also have that f is injective. (We note here that f is not necessarily a computable point of $C(\mathcal{X}; \mathcal{H})$ as s_e is not computable from e . In fact f should not be computable if $\mathcal{X}$ is computably compact.)

Lemma 5.4. *For each $s \in \omega$ f_s is continuous.*

Proof. We proceed via induction on the statement: “There exists M_s such that for any $x, y \in \mathcal{H}$, $\|f_s(x) - f_s(y)\|_{\mathcal{H}} \leq M_s \|x - y\|_{\mathcal{H}}$.” The base case is trivial as f_0 is the identity function. Suppose inductively that the statement holds for some $s \in \omega$. Evidently, if no requirement acts during stage s , then the statement similarly holds for $s+1$. We may thus assume that some requirement, say R_e , acts during stage s .

For convenience of notation, let $c(x) = \max\{0, 1 - \varepsilon^{-1} \|f_s(x) - f_s(x_s)\|_{\mathcal{H}}\}$, where ε is chosen as in the construction. Following the definition of f_{s+1} , we have that

$$\|f_{s+1}(x) - f_{s+1}(y)\|_{\mathcal{H}}^2 = \|f_s(x) - f_s(y)\|_{\mathcal{H}}^2 + (2^{-2e-1} |c(x) - c(y)|)^2.$$

It thus suffices to show that there exists k such that $|c(x) - c(y)| \leq k \|x - y\|_{\mathcal{H}}$ for any $x, y \in \mathcal{X}$. Consider the following cases.

Case 1: If $c(x) = c(y) = 0$, then we are done.

Case 2: Without loss of generality, suppose that $c(x) = 0$ and $c(y) > 0$. Then, $|c(x) - c(y)| = 1 - \varepsilon^{-1} \|f_s(y) - f_s(x_s)\|_{\mathcal{H}} > 0$. Since $c(x) = 0$, it must be that $\|f_s(x) - f_s(x_s)\|_{\mathcal{H}} \geq \varepsilon$. We thus obtain that

$$\begin{aligned} |c(x) - c(y)| &= \varepsilon^{-1} (\varepsilon - \|f_s(y) - f_s(x_s)\|_{\mathcal{H}}) \\ &\leq \varepsilon^{-1} |\|f_s(x) - f_s(x_s)\|_{\mathcal{H}} - \|f_s(y) - f_s(x_s)\|_{\mathcal{H}}| \\ &\leq \varepsilon^{-1} \|f_s(x) - f_s(x_s) - f_s(y) + f_s(x_s)\|_{\mathcal{H}} \\ &\leq \varepsilon^{-1} M_s \|x - y\|_{\mathcal{H}}. \end{aligned}$$

Case 3: If $c(x), c(y) > 0$, then $|c(x) - c(y)| = \varepsilon^{-1} |\|f_s(x) - f_s(x_s)\|_{\mathcal{H}} - \|f_s(y) - f_s(x_s)\|_{\mathcal{H}}|$. Applying the argument above allows us to also obtain that $|c(x) - c(y)| \leq \varepsilon^{-1} M_s \|x - y\|_{\mathcal{H}}$.

Thus, there exists M_{s+1} such that for any $x, y \in \mathcal{X}$, $\|f_{s+1}(x) - f_{s+1}(y)\|_{\mathcal{H}} \leq M_{s+1} \|x - y\|_{\mathcal{H}}$. $\square$

6. SPACES WITH A CONNECTED BASE

In this section, we consider spaces with a *connected presentation*, a computable Polish presentation where each basic open ball is connected.

Theorem 6.1. *If $\mathcal{X}$ has a connected presentation, then $\mathcal{X}$ is not computably categorical.*

Let $(\{\alpha_n\}_{n \in \mathbb{N}}, d_{\mathcal{X}})$ be a connected presentation of $\mathcal{X}$. We construct $\mathcal{Y} = \overline{(\{\beta_n\}_{n \in \mathbb{N}}, d_{\mathcal{Y}})}$ a computable Polish presentation of $\mathcal{X}$ satisfying the following requirements:

$\mathcal{R}_e$: Φ_e is not a homeomorphism from $\mathcal{Y}$ to $\mathcal{X}$.

The basic idea for each requirement is as follows. Initially keep $\mathcal{Y} = \mathcal{X}$, i.e., for each m, n , define $d_{\mathcal{Y}}(\beta_m, \beta_n) = d_{\mathcal{X}}(\alpha_m, \alpha_n)$. Then pick some ball $B(\beta_i, r)$ and search for $B(\alpha_j, q)$ satisfying both of the following.

- (1) $B(\alpha_j, q) \subseteq \Phi_e(B(\beta_i, r))$.
- (2) There exists $B(\beta_k, p)$ such that $d_{\mathcal{Y}}(\beta_i, \beta_k) < p$ and $B(\beta_k, p) \subseteq \Phi_e^{-1}(B(\alpha_j, q))$.

It is not hard to see that $B(\alpha_j, q)$ must exist if Φ_e is a homeomorphism. Additionally, since Φ_e is a computable homeomorphism, $B(\alpha_j, q)$ and $B(\beta_k, p)$ can be found effectively.

Once such a $B(\beta_k, p)$ is discovered, we ‘distort’ the space $\mathcal{Y}$ so that $\Phi_e^{-1}(B(\alpha_j, q))$ is not connected. This ensures that Φ_e fails as a homeomorphism since $B(\alpha_j, q)$ is connected. To do so, for each l such that $d_{\mathcal{X}}(\alpha_l, \alpha_k) = \frac{\varepsilon}{2}$ for some $\varepsilon < p$ (to be chosen), we define $d_{\mathcal{Y}}(\beta_l, \beta_k) > r$, placing β_l in a ‘fresh dimension’, and scaling the deformation to the other points β_l where l is such that $d_{\mathcal{X}}(\alpha_l, \alpha_k) < p$.

Roughly speaking, the procedure above stretches some sphere of radius $\frac{\varepsilon}{2}$ around α_k in $\mathcal{X}$ and places all such points ‘far away’ from α_i and α_k to obtain the relative positions of β_i, β_k , and β_l in $\mathcal{Y}$. Thus, at least intuitively, $\Phi_e^{-1}(B(\alpha_j, q))$ cannot possibly be connected as it must cover all such β_l that are now placed outside of $B(\beta_i, r)$ in $\mathcal{Y}$, but we have that $B(\beta_k, p) \subseteq \Phi_e^{-1}(B(\alpha_j, q)) \subseteq B(\beta_i, r)$ by (1) and (2) above.

Observe that the strategy for $\mathcal{R}_e$ ‘acts’ only on some local neighbourhood, and can thus be kept ‘sufficiently disjoint’ by careful choices of $r \in \mathbb{Q}^+$. We now give the details.

6.1. Construction. In order to keep the strategies of each $\mathcal{R}_e$ ‘disjoint’ we arrange our construction so that the neighbourhoods each $\mathcal{R}_e$ deforms are mutually disjoint. To arrange for this, non-uniformly fix the diameter of $\mathcal{X}$, $D = \sup\{d_{\mathcal{X}}(x, y) \mid x, y \in \mathcal{X}\}$. If $D \leq 1$, by defining $d_{\mathcal{X}}^* = D^{-1}d_{\mathcal{X}}$, we may ensure that $\sup\{d_{\mathcal{X}}^*(x, y) \mid x, y \in \mathcal{X}\} \geq 1$. Additionally, we have that $(\{\alpha_n\}_{n \in \mathbb{N}}, d_{\mathcal{X}}) \cong (\{\alpha_n\}_{n \in \mathbb{N}}, d_{\mathcal{X}}^*)$. By ‘restricting’ each $\mathcal{R}_e$ to act only on a ball with radius $\leq 2^{-e-2}$ (with respect to $d_{\mathcal{X}}^*$), we may keep the balls for each $\mathcal{R}_e$ mutually disjoint while ensuring that there is sufficient space to act for every requirement. Thus, if $D < \infty$, we may assume without loss of generality that $D \geq 1$ by letting $d_{\mathcal{X}} = d_{\mathcal{X}}^*$. On the other hand, if $D = \infty$, then regardless of the choice of radii, we must still be able to keep the balls for each $\mathcal{R}_e$ disjoint. In particular, it suffices to ensure that the ball for $\mathcal{R}_e$ has radius $\leq 2^{-e-2}$.

At each stage s of the construction, our task is to define $d_{\mathcal{Y}}(\beta_i, \beta_j)[s]$ for each $i, j < s$, and to ensure that $\mathcal{Y} = \overline{(\{\beta_i\}_{i \in \mathbb{N}}, d_{\mathcal{Y}})} \cong \mathcal{X}$. To aid in defining $d_{\mathcal{Y}}[s]$, we shall define $f_s : \{\alpha_i\}_{i \in \mathbb{N}} \rightarrow \mathbb{Q}^{\geq 0}$. Let f_0 be the constant 0 function. We may also assume that at stage s of the construction, $d_{\mathcal{X}}(\alpha_i, \alpha_j)[s] \downarrow$ for each $i, j < s$. As discussed earlier, each requirement acts only on some ball with a sufficiently small radius based on their index that is disjoint from the balls chosen for all earlier requirements. We recursively pick indices i_e and radii $r_e \leq 2^{-e-2}$ so that the balls $B(\beta_{i_e}, r_e)$ (in $\mathcal{Y}$) and the balls $B(\alpha_{i_e}, r_e)$ (in $\mathcal{X}$) are pairwise disjoint.

- Pick $i_0 = 0$ and $r_0 = 2^{-2}$.
- Suppose that $i_{e'}$ and $r_{e'}$ has been defined for each $e' < e$. Search for the least j such that for each $e' < e$, $d_{\mathcal{X}}(\alpha_j, \alpha_{i_{e'}})[s] > r_{e'} + 2^{-s}$. Note that this implies that $\lim_s d_{\mathcal{X}}(\alpha_j, \alpha_{i_{e'}})[s] > r_{e'}$. Then define $i_e = j$ and let

$$r_e = \min(\{2^{-e-2}\} \cup \{r_{e'} + 2^{-s} - d_{\mathcal{X}}(\alpha_j, \alpha_{i_{e'}})[s] \mid e' < e\}) > 0.$$

In particular, note that $B(\beta_{i_e}, r_e) \cap B(\beta_{i_{e'}}, r_{e'}) = \emptyset$ for any $e' < e$.

At stage $s+1$, let $e < s$ be the least such that i_e, r_e have both been defined, and such that there exists $p, q \in \mathbb{Q}^+$ and $j, k \in \mathbb{N}$ satisfying the following.

- (1) $B(\alpha_j, q) \subseteq \Phi_e(B(\beta_{i_e}, r_e))$, and
- (2) $B(\beta_k, p) \subseteq \Phi_e^{-1}(B(\alpha_j, q))$, and
- (3) $d_{\mathcal{X}}(\alpha_k, \alpha_{i_e})[s+1] < p - 2^{-s-1}$.

Fix $k \in \mathbb{N}$ and $p \in \mathbb{Q}^+$ witnessing the above conditions. Additionally, pick $\varepsilon \in \mathbb{Q}^+$ so that $\varepsilon < p$, and $\varepsilon = \min\{d_{\mathcal{X}}(\alpha_i, \alpha_j)[s+1] - 2^{-s-1} \mid i, j < s+1\}$. By arranging the enumeration of $\mathcal{X}$ so that a new point α_n is enumerated only at a stage t when $d_{\mathcal{X}}(\alpha_n, \alpha_i)[t] > 2^{-t}$, we may assume that for each $i, j < s+1$, we always have that $d_{\mathcal{X}}(\alpha_i, \alpha_j)[s] - 2^{-s-1} > 0$, and thus, $\varepsilon > 0$. In particular, the ball $B(\alpha_k, \varepsilon)$ is currently empty. Now define f_{s+1} as follows:

$$f_{s+1}(\alpha_n)[t] = \begin{cases} f_s(\alpha_n)[t] & \text{if } d_{\mathcal{X}}(\alpha_n, \alpha_k)[t] \geq \varepsilon + 2^{-t}, \\ 2^{-e} \left(1 - \left(\frac{\varepsilon}{2}\right)^{-1} \left|d_{\mathcal{X}}(\alpha_n, \alpha_k)[t] - \frac{\varepsilon}{2}\right|\right) & \text{otherwise.} \end{cases}$$

If no such e exists, then let $f_{s+1} = f_s$. In either case, define for each $i, j < s+1$,

$$d_{\mathcal{Y}}(\beta_i, \beta_j)[s+1] = d_{\mathcal{X}}(\alpha_i, \alpha_j)[s+1] + |f_{s+1}(\alpha_i)[s+1] - f_{s+1}(\alpha_j)[s+1]|.$$

6.2. Verification. We now verify that $d_{\mathcal{Y}}$ as defined is a metric, and that $\mathcal{Y} \cong \mathcal{X}$.

Lemma 6.2. $f = \lim_s f_s$ exists.

Proof. Fix an arbitrary index n . From the definition of f_s , the value of $f_s(\beta_n)$ depends entirely on whether α_n is contained within some ball $B(\alpha_k, \varepsilon) \subseteq B(\alpha_{i_e}, r_e)$ (in fact $B(\alpha_k, \varepsilon)$ is formally contained in $B(\alpha_{i_e}, r_e)$), where $k \in \mathbb{N}$ and $\varepsilon \in \mathbb{Q}^+$ are chosen as in the construction. If $\alpha_n \notin B(\alpha_{i_e}, r_e)$ for any e , then it is evident that $\lim_s f_s(\alpha_n) = 0$. We may thus suppose that α_n is contained in some $B(\alpha_{i_e}, r_e)$.

From the choice of i_e and r_e , it follows that there is at most one e for which $\alpha_n \in B(\alpha_{i_e}, r_e)$. In addition, if $\alpha_n \in B(\alpha_{i_e}, r_e)$ is also contained in $B(\alpha_k, \varepsilon)$, then there exists some stage s so that for all $t \geq s$, $d_{\mathcal{X}}(\alpha_n, \alpha_k)[t] < \varepsilon - 2^{-t}$. That is, $\lim_s f_s(\alpha_n) = 2^{-e} \left(1 - \left(\frac{\varepsilon}{2}\right)^{-1} \left|\lim_s d_{\mathcal{X}}(\alpha_n, \alpha_k)[s] - \frac{\varepsilon}{2}\right|\right)$ which exists as $\lim_s d_{\mathcal{X}}(\alpha_n, \alpha_k)[s]$ must exist.

Now consider the case where $\alpha_n \in B(\alpha_{i_e}, r_e)$ for some e but $\alpha_n \notin B(\alpha_k, \varepsilon)$ (where k, ε are chosen based on e). We claim that $\lim_s f_s(\alpha_n) = 0$ for such an n . Since $\alpha_n \notin B(\alpha_k, \varepsilon)$, we have that $d_{\mathcal{X}}(\alpha_n, \alpha_k) \geq \varepsilon$. That is, at each stage t , $d_{\mathcal{X}}(\alpha_n, \alpha_k)[t] \geq \varepsilon - 2^{-t}$. Since $\alpha_n \in B(\alpha_{i_e}, r_e)$, there must be some stage s such that for all $t \geq s$, $d_{\mathcal{X}}(\alpha_n, \alpha_{i_e})[t] < r_e - 2^{-t}$. In particular, $d_{\mathcal{X}}(\alpha_n, \alpha_{i_{e'}})[t] \geq r_{e'} + 2^{-t}$ for all $t \geq s$ and all $e' \neq e$. Thus, for all $t \geq s$, we have that either $f_t(\alpha_n) = f_0(\alpha_n) = 0$, if $d_{\mathcal{X}}(\alpha_n, \alpha_k) \geq \varepsilon + 2^{-t}$, or $f_t(\alpha_n) = 2^{-e} \left(1 - \left(\frac{\varepsilon}{2}\right)^{-1} \left|d_{\mathcal{X}}(\alpha_n, \alpha_k)[t] - \frac{\varepsilon}{2}\right|\right)$ otherwise. By the assumption that $d_{\mathcal{X}}(\alpha_n, \alpha_k)[t] \geq \varepsilon - 2^{-t}$ for all t , we have that for sufficiently large t , $|d_{\mathcal{X}}(\alpha_n, \alpha_k)[t] - \frac{\varepsilon}{2}| \geq \varepsilon - 2^{-t} - \frac{\varepsilon}{2} = \frac{\varepsilon}{2} - 2^{-t} > 0$. This implies that for sufficiently large t , either $f_t(\alpha_n) = 0$ or $f_t(\alpha_n) \leq 2^{-e} \left(1 - \left(\frac{\varepsilon}{2}\right)^{-1} \left(\frac{\varepsilon}{2} - 2^{-t}\right)\right) = 2^{-e} \left(\frac{\varepsilon}{2}\right)^{-1} (2^{-t})$. Clearly, the latter approaches 0 as t tends to infinity, and thus, for n such that $\alpha_n \in B(\alpha_{i_e}, r_e)$ but $\alpha_n \notin B(\alpha_k, \varepsilon)$, $\lim_s f_s(\alpha_n) = 0$. $\square$

Lemma 6.3. $d_{\mathcal{Y}} = \lim_s d_{\mathcal{Y}}[s]$ exists, and is a computable metric.

Proof. We first verify that $d_{\mathcal{Y}}$ exists and is a metric. Applying Lemma 6.2 allows one to conclude that for any $i, j \in \mathbb{N}$, $d_{\mathcal{Y}}(\beta_i, \beta_j) = \lim_s d_{\mathcal{Y}}(\beta_i, \beta_j)[s]$ exists. It is also easy to see that $d_{\mathcal{Y}}(\beta_i, \beta_j) = d_{\mathcal{Y}}(\beta_j, \beta_i)$ and is non-negative. Additionally, $i \neq j$ iff $d_{\mathcal{X}}(\alpha_i, \alpha_j) \neq 0$ iff $d_{\mathcal{Y}}(\beta_i, \beta_j) \neq 0$. It remains

to verify that d_Y satisfies the triangle inequality. Fix $i, j, k \in \mathbb{N}$ and consider the following.

$$\begin{aligned} d_Y(\beta_i, \beta_j) &= d_X(\alpha_i, \alpha_j) + |f(\beta_i) - f(\beta_j)| \\ &= d_X(\alpha_i, \alpha_j) + |f(\beta_i) - f(\beta_k) + f(\beta_k) - f(\beta_j)| \\ &\leq d_X(\alpha_i, \alpha_k) + d_X(\alpha_k, \alpha_j) + |f(\beta_i) - f(\beta_k)| + |f(\beta_k) - f(\beta_j)| \\ &= d_Y(\beta_i, \beta_k) + d_Y(\beta_k, \beta_j). \end{aligned}$$

Thus, d_Y is a metric. Finally, since $f(\beta_i)$ is a uniformly computable real for all $i \in \mathbb{N}$, it follows that d_Y is a computable metric. $\square$

From the definition of f , it is also easy to see that f is continuous w.r.t. d_X . f may then be extended to be defined on $\mathcal{X}$ by taking $f(x) = \lim_n f(\alpha_{l_n})$ for any sequence $\{\alpha_{l_n}\}_{n \in \mathbb{N}}$ converging to x . In a similar way, the ‘identity map’ $\alpha_i \mapsto \beta_i$ map be extended to a bijection from $\mathcal{X} \rightarrow \mathcal{Y}$. It remains to argue that this map is a homeomorphism.

Lemma 6.4. *The (identity map) $h : \mathcal{X} \rightarrow \mathcal{Y}$ defined by extending the map $\alpha_i \mapsto \beta_i$ is a homeomorphism.*

Proof. By the definition of d_Y , we have that for any $a, b \in \mathcal{X}$, $d_X(a, b) \leq d_Y(a, b)$. Thus, for any open ball $B(\alpha_i, r) \subseteq \mathcal{X}$, the ball $B(\beta_i, r) \subseteq \mathcal{Y}$ is a subset of $B(\alpha_i, r)$. Now, given $B(\beta_i, r)$, we want to find $0 < q$ (necessarily bounded by r) so that $B(\alpha_i, q) \subseteq B(\beta_i, r)$. In particular, we require q to be such that for any $x \in B(\alpha_i, q)$, $|f(x) - f(\alpha_i)| < r - q$. But such a q necessarily exists, since $|f(x) - f(\alpha_i)|$ tends to 0 as q tends to 0, while $r - q$ tends to $r > 0$. Thus, the topology generated by d_X and d_Y are equivalent. $\square$

Lemma 6.5. *Each requirement $\mathcal{R}_e$ is satisfied.*

Proof. From the construction, it follows that either there exists some stage s such that $e < s$ is the least for which i_e, r_e have been defined and satisfying conditions (1), (2), (3), or $\mathcal{R}_e$ is trivially satisfied. On the other hand, if we do find $p, q \in \mathbb{Q}^+$ satisfying (1), (2), and (3) for $\mathcal{R}_e$ as stated in the construction, then we claim that Φ_e fails to witness a homeomorphism from $\mathcal{Y} \rightarrow \mathcal{X}$.

Conditions (1) and (2) guarantee that $\Phi_e(B(\beta_k, p)) \subseteq B(\alpha_j, q) \subseteq \Phi_e(B(\beta_{i_e}, r_e))$. By the assumption that $\mathcal{X}$ is a connected presentation, $B(\alpha_j, q)$ must be connected. However all points x where $d_X(x, \alpha_k) = \frac{\varepsilon}{2}$ are mapped to points y where $d_Y(y, \beta_k) = 2^{-e} > r_e$. In particular, $y \notin B(\beta_{i_e}, r_e)$. Thus, $B(\beta_k, p)$ cannot be connected and the preimage of $B(\alpha_j, q)$ also cannot be extended into a connected set without using balls not contained in $B(\beta_{i_e}, r_e)$. $\square$

7. THE CLOSED UNIT INTERVAL

In this subsection we will investigate the topological categoricity of the closed unit interval. We let $[0, 1]$ denote the standard presentation of the closed unit interval. That is, each rational $0 \leq r \leq 1$ is a special point, and $d(x, y) = |x - y|$. Obviously if a, b are left-c.e. and right-c.e. reals respectively, then $[a, b]$ is an effective presentation of $[0, 1]$. Furthermore any homeomorphism $f : [0, 1] \rightarrow [a, b]$ must satisfy $f(0) = a$ or $f(0) = b$, otherwise $f([0, 1]) = [a, b] - \{f(0)\}$ would be disconnected. Therefore, if a or b is not computable then $[a, b]$ and $[0, 1]$ are not effectively homeomorphic, and so $[0, 1]$ is not computably categorical. This fact is also a direct consequence of Theorem 5.1 and/or of Theorem 6.1. We prove more.

Theorem 7.1. *The degree of categoricity for the closed unit interval $[0, 1]$ is $0''$.*

Proof. The first task is to establish $\mathbf{0}''$ as an upper bound on categoricity. To see this, define the relation $BTW(x, y, z)$ for distinct $x, y, z \in \mathcal{M} \cong [0, 1]$ if for all $\delta > 0$, there is a positive $\varepsilon < \delta$ such that for all ε -chains C from x to z , $d(y, C) < \delta$. Intuitively, this says that any path from x to z in $\mathcal{M}$ should contain y . As shown in [HTH], $BTW(x, y, z)$ is Δ_3^0 in the names of x, y, z , provided they are distinct points. Together with the fact that any homeomorphism between two copies of $[0, 1]$ must preserve this relation, it is not hard to see that $[0, 1]$ is $\mathbf{0}''$ -categorical.

Now we prove that this result is sharp. For that, we need to construct a computable copy, $(\{\alpha_n\}_{n \in \omega}, d)$, of $[0, 1]$ such that $\emptyset''$ is computable from any code for a homeomorphism between this copy and the usual copy of $[0, 1]$.

We build our copy of $[0, 1]$ as the union of ω -many segments. Each segment will have a start point and an end point that is the start point of the next segment. To encode $\emptyset''$ we will use two special points, called a and b , that lie between the start and end points. The ordering of a and b in segment e will encode if φ_e is total or not, by having $a < b$ if and only if φ_e is total.

We will build each segment as a subset of $[-1, 1]^\omega$ with the usual metric, by defining a dense set of points $\{\alpha_n\}_{n \in \omega}$. We define the start point as $(-1) \smallfrown 0^\omega$ and the end point as $1 \smallfrown 0^\omega$. We define $a = \frac{-1}{2} \smallfrown 0^\omega$ and $b = \frac{1}{2} \smallfrown 0^\omega$.

The idea here is that if φ_e is total, then the closure of $\{\alpha_n\}_{n \in \omega}$ is a fractal homeomorphic to $[0, 1]$. Furthermore we define an ordering on the points in this fractal by letting $\alpha <_0 \beta \iff \alpha(0) < \beta(0)$, and claim that the inherited topology from $[-1, 1]^\omega$ is the same as the interval topology with respect to $<_0$.

If, on the other hand, $\varphi_e(n) \uparrow$, then the construction will only use finitely many dimensions of $[-1, 1]^\omega$ and we will end up with a piecewise linear path from the start point to b to a to the end point rather than a fractal.

At stage $s + 1$ we will have chosen $\alpha_0, \dots, \alpha_{n_s}$, and will have assigned an order to these points, $<_s$. We define the cycle at stage s , $\text{cycle}(s)$, to be the least c such that $\varphi_{e,s}(c) \uparrow$. This will determine the number of dimensions we are using and the current path we are constructing. We will also have defined a piecewise linear path, called $\text{path}(c)$, that lives in $[-1, 1]^{c+1}$. We will have ensured that when projected onto $[-1, 1]^{\text{cycle}(s)+1}$ all points lie on $\text{path}(s)$, and we must ensure that all future points also project onto this path.

The construction goes as follows. If at stage $s + 1$ we observe $c = \text{cycle}(s + 1) \neq \text{cycle}(s)$ then it is time for us to determine a new order and start adding points towards the construction of a new path in $c + 2$ dimensions. The order $<_{s+1}$ will be as follows:

- If $\alpha_i <_0 \alpha_j <_0 a$ then $\alpha_i <_{s+1} \alpha_j <_{s+1} b$.
- If $a <_0 \alpha_i <_0 \alpha_j <_0 b$ then $b <_{s+1} \alpha_j <_{s+1} \alpha_i <_{s+1} a$ (reverse of $<_0$).
- If $b <_0 \alpha_i <_0 \alpha_j$ then $a < \alpha_i <_{s+1} \alpha_j$.

We will define the new $\text{path}(c)$ to be the path consisting of line segments between adjacent points in $\alpha_0, \dots, \alpha_{n_s}$ under the $<_0$ order. This will mean that the order of points on $\text{path}(c)$ matches up with the order according to $<_0$. Since all these points project onto $\text{path}(\text{cycle}(s))$, $\text{path}(c)$ will also project onto the previous path.

Finally, at this stage we add two more points to our sequence, r_a and r_b . r_b is the predecessor of b and r_a is the successor of a in $<_{s+1}$. We choose r_a to be the unique point in $[-1, 1]^\omega$ that projects onto $\text{path}(c)$, has $r_a(0) = a(0) + 1/p_{2s}$, $r_a(c + 1) = 1$ and $r_a(k) = 0$ for all other unspecified values (here p_{2s} is considered to be the $2s$ -th prime number. This is used to avoid having $\alpha_i(0) = \alpha_j(0)$ for any future $i > j > n_s$). Similarly we define $r_b(0) = b(0) - 1/p_{2s+1}$ and $r_b(c + 1) = -1$. This

means that at future stages we add points along a path that looks something like the one in Figure 1

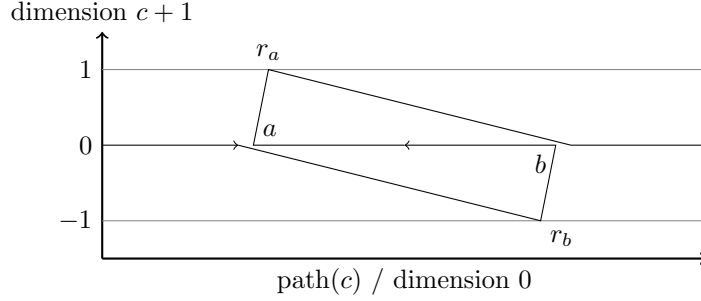


FIGURE 1. the space we are building while $\text{cycle}(s) = c$.

If at stage s we instead observe $c = \text{cycle}(s + 1) = \text{cycle}(s)$ then we density: for each adjacent pair $\alpha_i <_s \alpha_j$ we add a point q in the middle with $q(k) = \frac{\alpha_i(k) + \alpha_j(k)}{2}$ for $k = 0, c + 1$ and the other coordinates chosen so that q projects onto $\text{path}(c)$.

This completes the construction.

For the verification, it is enough for us to observe that our early claims are met, that is, we must interval is indeed homeomorphic to $[0, 1]$ and have $a < b$ if and only if φ_e is total.

If we are in the case that $\varphi_e(c) \uparrow$ for some least c , Then we will have a piecewise linear path that looks something like the one in Figure 1. In this case we will have $b < a$ as desired, and being piecewise linear, we can see that it is indeed homeomorphic to $[0, 1]$.

If on the other hand, we have that φ_e is total, then we want to argue that the order topology under $<_0$ matches up with the topology inherited from $[-1, 1]^\omega$. This will give us that $a < b$ and, as a compact Polish space and dense linear order, this means our space must be homeomorphic to $[0, 1]$.

Consider some points $\alpha_i <_0 \alpha_j$. Since $\{f \in [-1, 1]^\omega : \alpha_i(0) < f(0) < \alpha_j(0)\}$ is an open set in $[-1, 1]^\omega$, it must be that the interval (α_i, α_j) is open under the inherited topology.

Consider some arbitrary ball, $B(\alpha_k, q)$. To show that this is open under the order topology it is enough for us to show that we can always find some interval $(\alpha_i, \alpha_j) \subseteq B(\alpha_k, q)$ such that $\alpha_k \in (\alpha_i, \alpha_j)$. To do this, let c be large enough that $2^{-c} < \frac{q}{2}$. Consider $\text{path}(c)$. This path lies in $[-1, 1]^{c+1}$, so any point that projects onto $\text{path}(c)$ must be within 2^{-c} distance of the path. Since $<_0$ is dense and $\text{path}(c) \cong [0, 1]$, we can choose α_i, α_j such that $\alpha_k \in (\alpha_i, \alpha_j)$ and the part of $\text{path}(c)$ that lies between the projections of α_i and α_j has diameter $< \frac{q}{2}$. So if we have some point $\alpha \in (\alpha_i, \alpha_j)$, then $d(\alpha, \alpha_k) \leq \frac{q}{2} + \frac{q}{2} = q$ as desired. $\square$

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